NHL STENDEN university of applied sciences

COMPUTER SCIENCE

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Introduction

My name is Francisco Ponte Pimentel, and this second-period portfolio builds upon the foundations established during the first stage of my studies at NHL Stenden University. While the previous portfolio focused on introductory work in Computer Science and Embedded Systems, this new period marks a shift toward more specialised and technically structured projects.

In Computer Science, I am currently developing my skills on Python. Alongside Python, I am progressively working with SQL, HTML, CSS, JavaScript, and Flask to understand how software systems are designed, implemented, and connected across different layers of development. These technologies allow me to transition from conceptual understanding to the construction of practical, interactive, and well-structured applications.

Simultaneously, my work in Embedded Systems advances through the ongoing development of the Relay Bot, a project that challenges me to combine hardware control, efficient programming, and systematic testing. This hands-on experience strengthens my understanding of how physical components and digital logic intersect to form functional systems.

Throughout this period, I continue to apply the same mindset that has guided my academic journey so far: curiosity supported by discipline, attention to detail, and a commitment to understanding not only how technologies work, but why they are designed in specific ways. I value adaptability, clear communication, and collaborative learning, as these qualities allow me to engage with increasingly complex challenges.

Computer Science

Week 1

Hello, Again

```python
import cs50                                          # Get access to get_string
name = cs50.get_string("What is your name? ")        # Asks to input a name
print(f"hello, {name}")                              # Print "hello, + name"
```

Hello, Again – Result

Figure 1- Hello, Again | Result

me50 / users / franciscpime / cs50 / problems / 2025 / x / sentimental / hello

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Mario Less

```python
import cs50                                          # Get acces to get_int
while True:
    n = cs50.get_int("Height: ")                     # Asks to input a number
    if n >= 1 and n <= 8:                            # Should be between [1, 8]
        break                                        # Get out of the loop
    else:
        print("Number out of range. Try again.")     # Print to try again
# Draw the pyramid
for i in range(n):                                   # E.g: n = 4; i from 0 to 3
    num_spaces = n - i - 1                           # 4 - 0 - 1 = 3 spaces
    num_hashes = i + 1                               # 0 + 1 = 1 hashe
    for space in range(num_spaces):                  # Print spaces until i = 2 (0 to 2)
        print(" ", end="")                           # Print space
    for hash in range(num_hashes):                   # i = 3; num_hashes = 4
        print("#", end="")                           # Print hash
    print()                                          # New line
```

Mario Less – Result

Figure 2 - Mario Less | Result

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Mario More

```python
from cs50 import get_int                        # Get access to get_int
while True:
    n = get_int("Height: ")                     # Asks to input a number
    if 1 <= n <= 8:                             # Should be between [1, 8]
        break                                   # Get out of the loop
    else:
        print("Number out of range. Try again.")  # Print to try again
# Draw 2 pyramids
for i in range(n):                              # E.g: n = 5; i from 0 to 4
    num_spaces = n - i - 1                      # 5 - 0 - 1 = 4 spaces
    num_hashes = i + 1                          # 0 + 1 = 1 hashe
    # 1st Pyramid
    for space in range(num_spaces):             # Print spaces until i = 3 (0 to 3)
        print(" ", end="")                      # Print space
    for hash in range(num_hashes):              # i = 4; num_hashes = 5
        print("#", end="")                      # Print hash
    print("  ", end="")                         # Print spaces between pyramids
    # 2nd Pyramid
    for hash in range(num_hashes):              # i = 4; num_hashes = 5
        print("#", end="")                      # Print hash
    print()                                     # New line
```

Mario More – Result

Figure 3 - Mario More | Result

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Cash

```python
from cs50 import get_float                  # Get acces to get_float
while True:
    dollars = get_float("Change: ")         # Asks to input a number
    if dollars >= 0:                        # If the number is positive
        break                               # Get out of the loop
    else:
        print("Print a positive number.")   # Else, print until print a positive number
cents = round(dollars * 100)                # Convert dollars to cents
coins = 0                                   # Total number of coins
coin = 25                                   # Coins available (in cents): 25, 10, 5 e 1
while cents > 0:                            # Continue until all change has been given
    if cents >= coin:                       # If I can still use the current coin.
        cents -= coin                       # Do cents - 25
        coins += 1                          # Add 1 coin
    elif coin == 25:                        # Coin that was 25
        coin = 10                           # Now is 10
    elif coin == 10:                        # Coin that was 10
        coin = 5                            # Now is 5
    elif coin == 5:                         # Coin that was 5
        coin = 1                            # Now is 1
print(f"{coins}")                           # Print the result
```

Cash – Result

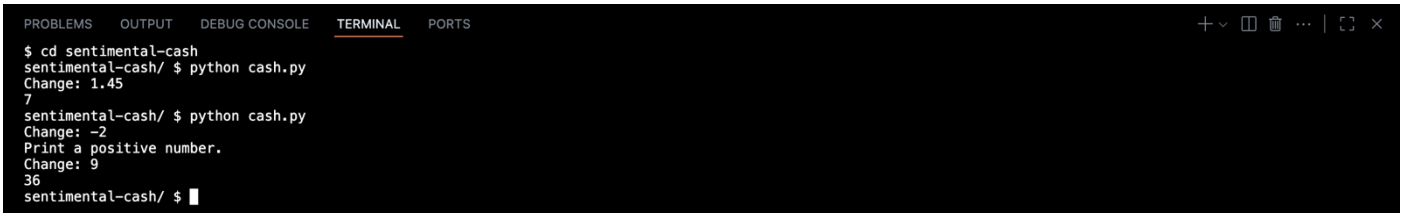

Figure 4 - Cash | Result

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Credit

```
from cs50 import get_float                  # Get acces to get_float
while True:
    dollars = get_float("Change: ")         # Asks to input a number
    if dollars >= 0:                        # If the number is positive
        break                               # Get out of the loop
    else:
        print("Print a positive number.")   # Else, print until print a positive number
cents = round(dollars * 100)                # Convert dollars to cents
coins = 0                                   # Total number of coins
coin = 25                                   # Coins available (in cents): 25, 10, 5 e 1
while cents > 0:                            # Continue until all change has been given
    if cents >= coin:                       # If I can still use the current coin.
        cents -= coin                       # Do cents - 25
        coins += 1                          # Add 1 coin
    elif coin == 25:                        # Coin that was 25
        coin = 10                           # Now is 10
    elif coin == 10:                        # Coin that was 10
        coin = 5                            # Now is 5
    elif coin == 5:                         # Coin that was 5
        coin = 1                            # Now is 1
print(f"{coins}")                           # Print the result
```

Credit – Result

Figure 5 - Credit | Result

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Readability

```python
def main():
    # Asks to input a text
    text = input("Text: ")     # Example input: "One fish. Two fish. Red fish. Blue fish."
    # count_letters("One fish. Two fish. Red fish. Blue fish.") >> 29
    letters = count_letters(text)
    words = count_words(text)                       # count_words(...) >> 8
    sentences = count_sentences(text)               # count_sentences(...) >> 4
    # Avoids division by 0 = error if the text is empty, e.g: "" >> words = 0
    if words == 0:
        words = 1              # Without this, L = letters / words would cause an error
    # Coleman-Liau formula:
    # L = average number of letters per 100 words
    # S = average number of sentences per 100 words
    L = float(letters) / words * 100                # (29 / 8) * 100 = 362.5
    S = float(sentences) / words * 100              # (4 / 8) * 100 = 50.0
    # index = round(0.0588 * 362.5 - 0.296 * 50.0 - 15.8) = -9
    index = round(0.0588 * L - 0.296 * S - 15.8)
    # Final result
    if index < 1:
        print("Before Grade 1")         # For this example the result is: "Before Grade 1"
    elif index >= 16:
        print("Grade 16+")
    else:
        print(f"Grade {index}")
def count_letters(text):
    count = 0
    for c in text:
        if c.isalpha():          # It only counts letters (ignores spaces and punctuation)
            count += 1
    return count                 # Returns 29
def count_words(text):
    count = 1          # It starts at 1 because the number of words = number of spaces + 1
    for c in text:
        if c.isspace():          # Each space counts as a word separator
            count += 1
            # Example: 7 spaces >> 8 words
    return count                 # Returns 8
def count_sentences(text):
    count = 0
    for c in text:
        if c in ['.', '!', '?']: # Count each period, exclamation point, or question mark
            count += 1
    return count                 # Returns 4
main()
```

Readability - Result

Figure 6 - Readability | Result

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DNA

```
import csv
import sys
def main():
    # Check for command-line usage
    if len(sys.argv) != 3:       # Check if there are not exactly 3 command-line arguments
        print("Usage: python dna.py database.csv sequence.txt")     # Show correct usage
        return                                                      # Stop the program
    # Read database file into a variable
    with open(sys.argv[1], "r") as data_file:             # Open the first file (database)
        reader = csv.DictReader(data_file)   # Read the CSV file as a list of dictionaries
        file = [row for row in reader]       # Store all rows from the CSV into a list
    # Read DNA dequence file into a variable
    with open(sys.argv[2], "r") as txtfile:  # Open the second file (DNA sequence)
        dna_sequence = txtfile.read()  # Read the entire content of the file into a string
    # Find longest match of each STR in DNA sequence
    str_counts = {}                     # Create an empty dictionary to store STR counts
    for str_seq in file[0].keys():      # Loop through all keys in the first row of the CSV
        if str_seq == "name":           # Skip the "name" column
            continue
      # Find the longest run of the STR in the DNA sequence and store it in the dictionary
        str_counts[str_seq] = longest_match(dna_sequence, str_seq)
    # TODO: Check database for matching profiles
    for person in file:                           # Loop through each person in the database
        match = True                              # Assume a match until proven otherwise
        for str_seq in person.keys():             # Loop through each STR for the person
            if str_seq == "name":                 # Skip the "name" column
                continue
            # Compare the person's STR count with the one found in the DNA sequence
            if int(person[str_seq]) != str_counts[str_seq]:
                match = False               # If they don't match, set match to False
                break                       # No need to check further STRs for this person
        if match:                                                   # If all STRs matched
            print(person['name'])                                   # If all STRs matched
            return                                                  # Exit the function
    # Print this if no person's STR counts match the DNA sequence
    print("No match")
def longest_match(sequence, subsequence):
    """Returns length of longest run of subsequence in sequence."""
    #Inicialize variables
    longest_run = 0
    subsequence_length = len(subsequence)
    sequence_length = len(sequence)
    # Check each position in sequence for consecutive repeats of subsequence
    for i in range(sequence_length):
        #Inicialize count of consecutive runs
        count = 0
  # Check for a subsequence match in a "substring" within sequence
```

```
    # If a match, move substracting to next potential match in sequence
    # Continue moving substracting and checking for matches until out of consecutive matches
        while True:
            # Adjust substracting start and end
            start = i + count * subsequence_length
            end = start + subsequence_length
            # If there is a match in the substracting
            if sequence[start:end] == subsequence:
                count += 1
            else:
                break
        # Update longest run if current run is longer
        longest_run = max(longest_run, count)
    # After checking for runs at each character in sequence, return longest run found
    return longest_run
main()
```

DNA – Result

Figure 7 - DNA | Result

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Scrabble

```python
from cs50 import get_string                          # Get access to get_string
# Points for letters A to Z
POINTS = [1, 3, 3, 2, 1, 4, 2, 4, 1, 8, 5, 1, 3,     # Points for letters A to M
          1, 1, 3, 10, 1, 1, 1, 1, 4, 4, 8, 4, 10]   # Points for letters N to Z
def main():
    # Get input words from both players
    word1 = get_string("Player 1: ")                 # Ex.: Dog
    word2 = get_string("Player 2: ")                 # Ex.: Cat
    # Compute scores for both words
    score1 = compute_score(word1)                    # Dog >> 2 + 1 + 2 = 5
    score2 = compute_score(word2)                    # Cat >> 3 + 1 + 1 = 5
    # Compare the scores and print the result
    if (score1 > score2):
        print("Player1 wins!")
    elif (score1 < score2):
        print("Player2 wins!")
    else:
        print("Tie!")                                # Both have the same score
def compute_score(word):
    score = 0                                        # Initialize score to 0
    for char in word:                                # For each character in the word
        if char.isalpha():                           # Check if the character is a letter
            index = ord(char.lower()) - ord('a')     # Get the index (0 for a, 25 for z)
            score += POINTS[index]           # Add the corresponding points to the score
    return score                                     # Return the total score
main()                                       # Call the main function to start the program
```

Scrabble – Result

Figure 8 - Scrabble | Result

Caesar

```python
from cs50 import get_string                                  # Get access to get_string
import sys                                                   # Get access to sys.argv
def main(argc=None, argv=None):
    if (argc != 2):                                  # Check for correct number of arguments
        print("Usage: python caesar.py key")
        return 1
    if ((argv[1].isdigit()) == False):               # Check if key is a non-negative integer
        print("Key must be a non-negative integer.")
        return 1
    plaintext = get_string("plaintext: ")                    # Get plaintext from user
    key = int(argv[1])                                       # Get key from user
    ciphertext = encrypt(plaintext, key)                     # Encrypt the plaintext
    print(f"ciphertext: {ciphertext}")                       # Print the ciphertext
def encrypt(plaintext, key):
    ciphertext = ""                                  # Initialize ciphertext as an empty string
    for char in plaintext:                           # For each character in the plaintext
        if char.isupper():
            # Shift uppercase letters
            shifted = (ord(char) - ord('A') + key) % 26 + ord('A')
            ciphertext += chr(shifted)
        elif char.islower():
            # Shift lowercase letters
            shifted = (ord(char) - ord('a') + key) % 26 + ord('a')
            ciphertext += chr(shifted)
        else:
            ciphertext += char                   # Non-alphabetic characters remain unchanged
    return ciphertext
main(argc = len(sys.argv), argv = sys.argv)   # Call main with command-line arguments
```

Caesar – Result

Figure 9 - Caesar | Result

Plurality

```
from cs50 import get_string, get_int        # Get access to CS50 library functions
import sys                                  # Import sys module for command-line arguments
MAX = 9                                                 # Max of 9 candidates
candidates = []                                         # List to store candidates
def main(argc = None, argv = None):
    # Check for invalid usage (must provide at least one candidate)
    if argc < 2:
        print("Usage: plurality.py [candidate ...]") # Ex.: python plurality.py Ana Carlos
        return 1                                    # Exit with error code
    number_of_candidates = argc - 1                 # Number of candidates provided
    # Check if the number of candidates exceeds the maximum allowed
    if number_of_candidates > MAX:
        # Ex.: Maximum number of candidates is 9
        print(f"Maximum number of candidates is {MAX}")
        return 2                                            # Exit with error code
    # Populate the candidates list with dictionaries
    for i in range(1, argc):
        # Each candidate is stored as {"name": candidate_name, "votes": 0}
        candidates.append({"name": argv[i], "votes": 0})# Ex.: {"name": "Ana", "votes": 0}
    voter_count = get_int("Number of voters: ")         # Prompt user for number of voters
    # Ex.: 3 voters
    # Loop over all voters to collect votes
    for i in range(voter_count):
        name = get_string("Vote: ")                     # Prompt voter for their choice
        # Ex.: Vote: Ana
        if not vote(name):                              # Process the vote
            print("Invalid vote.\n")                    # Print message if vote is invalid
            # Ex.: Vote: Dave >> Invalid vote.
    print_winner()                                  # Display the winner(s) after all votes
# Function to process a vote
def vote(name):
    for candidate in candidates:                    # Loop through all candidates
        if candidate["name"] == name:               # Check if input matches candidate name
            candidate["votes"] += 1                 # Increment vote count for that candidate
            # Ex.: Ana votes += 1
            return True                             # Vote was successful
    return False                                    # Return False if no match found
# Function to print the winner(s)
def print_winner():
    max_votes = 0                                   # Initialize maximum votes to 0
    # Find the highest number of votes
    for candidate in candidates:                            # Loop through all candidates
        # Check if candidate has more votes than current max
        if candidate["votes"] > max_votes:
            max_votes = candidate["votes"]                  # Update max_votes
            # Ex.: max_votes = 2 if Ana has 2 votes
    # Print all candidates with maximum votes
    for candidate in candidates:
        if candidate["votes"] == max_votes:   # Check if candidate's votes equal max_votes
            print(candidate["name"])          # Print winner's name
            # Ex.: Ana
# Call main function with command-line arguments Ex.: python plurality.py Ana Carlos
main(argc=len(sys.argv), argv=sys.argv)
```

Plurality – Result

Figure 10 - Plurality | Result

Week 2

Songs

```
1.sql SELECT name
      FROM songs;
2.sql SELECT name
      FROM songs
      ORDER BY tempo ASC;
3.sql SELECT name
      FROM songs
      ORDER BY duration_ms
      DESC LIMIT 5;
4.sql SELECT name
      FROM songs
      WHERE danceability > 0.75
      AND energy > 0.75
      AND valence > 0.75;
5.sql SELECT AVG(energy)
      FROM songs;
6.sql SELECT name
      FROM songs
      WHERE artist_id
      IN (SELECT id FROM artists WHERE name = 'Post Malone');
7.sql SELECT AVG(energy)
      FROM songs
      WHERE artist_id
      IN (SELECT id FROM artists WHERE name = 'Drake');
8.sql SELECT name
      FROM songs
      WHERE name
      LIKE '%feat.%';
```

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tar.gz • zip

Movies

```
1.sql SELECT title
      FROM movies
      WHERE year = 2008;
2.sql SELECT birth
      FROM people
      WHERE name = 'Emma Stone';
3.sql SELECT title
      FROM movies
      WHERE year >= 2018
      ORDER BY title ASC;
4.sql SELECT COUNT(rating)
      FROM ratings
      WHERE rating = 10.0;
5.sql SELECT title, year
      FROM movies
      WHERE title
      LIKE 'Harry Potter%'
      ORDER BY year ASC;
6.sql SELECT AVG(rating)
      FROM ratings
      WHERE movie_id
      IN (SELECT id
      FROM movies
      WHERE year = 2012);
7.sql SELECT movies.title, ratings.rating
      FROM movies
      JOIN ratings
      ON movies.id = ratings.movie_id
      WHERE movies.year = 2010
      ORDER BY ratings.rating DESC, movies.title ASC;
8.sql SELECT people.name
      FROM people
      JOIN stars
      ON people.id = stars.person_id
      JOIN movies
      ON stars.movie_id = movies.id
      WHERE movies.title = 'Toy Story';
9.sql SELECT name
      FROM people
      JOIN stars
      ON person_id = people.id
      JOIN movies
      ON movies.id = stars.movie_id
      WHERE movies.year = 2004
      ORDER BY birth ASC;
10.sql SELECT name
       FROM people
       JOIN directors
       ON person_id = people.id
       JOIN ratings
       ON ratings.movie_id = directors.movie_id
       WHERE rating >= 9.0;
11.sql SELECT title
```

```
       FROM movies
       JOIN stars
       ON movies.id = stars.movie_id
       JOIN ratings
       ON ratings.movie_id = movies.id
       JOIN people
       ON people.id = stars.person_id
       WHERE people.name = 'Chadwick Boseman'
       ORDER BY rating DESC
       LIMIT 5;
12.sql SELECT title
       FROM movies
       JOIN  stars
       ON movies.id = stars.movie_id
       JOIN  people
       ON people.id = stars.person_id
       WHERE people.name
       IN ('Bradley Cooper', 'Jennifer Lawrence')
       GROUP BY movies.id
       HAVING COUNT(DISTINCT people.id) = 2;
13.sql SELECT DISTINCT P1.name
       FROM people AS P1
       JOIN stars AS S1
       ON P1.id = S1.person_id
       WHERE S1.movie_id
       IN (SELECT S2.movie_id
       FROM stars AS S2
       JOIN people AS P2
       ON P2.id = S2.person_id
       WHERE P2.name = 'Kevin Bacon'
       AND P2.birth = 1958)
       AND P1.id != (SELECT id
       FROM people
       WHERE name = 'Kevin Bacon'
       AND birth = 1958);
```

me50 / users / franciscpime / cs50 / problems / 2025 / x / movies

#1 submitted a few seconds ago, Sunday, November 16, 2025 11:50 AM CET
check50 14/14 • 0 comments
tar.gz • zip

```
14.sql INSERT INTO movies (id, title, year)
       VALUES
           (80684, "Star Wars: Episode V - The Empire Strikes Back", 1980),
           (86190, "Star Wars: Episode VI - Return of the Jedi", 1983),
           (120915, "Star Wars: Episode I - The Phantom Menace", 1999),
           (121765, "Star Wars: Episode II - Attack of the Clones", 2002),
           (121766, "Star Wars: Episode III - Revenge of the Sith", 2005);
15.sql INSERT INTO people (id, name, birth)
       VALUES
           (191, "Ewan Gordon McGregor", 1971),
           (204, "Natalie Portman", 1981),
           (159789, "Hayden Christensen", 1981),
           (184, "George Lucas", 1944);
16.sql INSERT INTO stars (movie_id, person_id)
       VALUES
           (121766, 191),
           (121766, 204),
           (121766, 159789);
       SELECT people.id, people.name
       FROM movies
       JOIN stars
       ON movies.id = stars.movie_id
       JOIN people
       ON stars.person_id = people.id
       WHERE title = 'Star Wars: Episode III - Revenge of the Sith'
       LIMIT 3;
17.sql SELECT temp.person_id, temp.movie_id
       FROM (
       SELECT person_id, movie_id
       FROM directors
       WHERE person_id
       IN (191, 204, 159789, 184)
       ) AS temp
       WHERE temp.movie_id
       IN (80684,86190,120915,121765,121766);
       SELECT title
       FROM movies
       JOIN directors
       ON movies.id = directors.movie_id
       JOIN people
       ON directors.person_id = people.id
       WHERE people.name ="George Lucas"
       AND movies.title
       LIKE 'Star Wars%';
18.sql SELECT title, rating
       FROM movies
       JOIN ratings
       ON movies.id = ratings.movie_id
       WHERE ratings.rating >= 8.0;
19.sql UPDATE ratings
       SET rating = 10.0
       WHERE movie_id = (
           SELECT id
```

```
            FROM movies
            WHERE title = 'Star Wars: Episode VI - Return of the Jedi';
        );
        SELECT rating
        FROM ratings
        WHERE movie_id = (
            SELECT id
            FROM movies
            WHERE title = 'Star Wars: Episode VI - Return of the Jedi'
        );
```

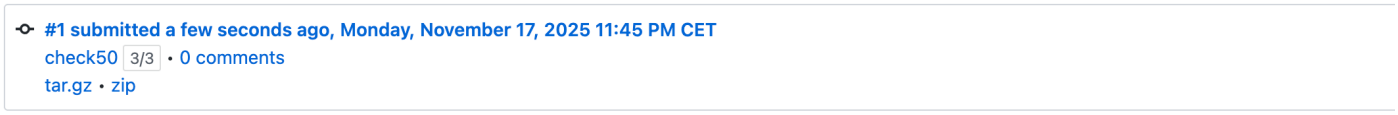
Fiftyville

```
-- Log of the queries I used in this investigation
-------------------------------------------------------------
-- 1. See what happened at the crime scene
-- Objective:
-- Let's see what happened on Humphrey Street that day.
-- Understand the crime and who was involved.
SELECT description
FROM crime_scene_reports
WHERE year = 2024
AND month = 7
AND day = 28
AND street = 'Humphrey Street';
-- Notes:
-- The CS50 duck was stolen at 10:15 at the bakery.
-- There are 3 witnesses who mentioned the bakery >> they are our initial lead.
-------------------------------------------------------------
-- 2. Read all the interviews from the day
-- Objective:
-- See what the witnesses said about the incident.
SELECT name, transcript
FROM interviews
WHERE year = 2024
AND month = 7
AND day = 28;
-- Notes:
-- Ruth >> saw the thief leave in 10 min. (10:15 - 10:25)
-- Eugene >> saw the thief earlier withdrawing money from the Leggett Street ATM.
-- Raymond >> heard the thief call (<60s) and talk about the earlier flight the next day.
-- 3 filters: time, ATM, and short call.
-------------------------------------------------------------
-- 3. Who left the bakery between 10:15 and 10:25?
-- Objective:
-- Confirm who was actually there at the right time to leave with the duck.
SELECT people.name
FROM people
JOIN bakery_security_logs
ON people.license_plate = bakery_security_logs.license_plate
WHERE year = 2024
AND month = 7
AND day = 28
AND hour = 10
```

```sql
AND minute BETWEEN 15 AND 25;
-- Notes:
-- Suspects: Vanessa, Bruce, Barry, Luca, Sofia, Iman, Diana, Kelsey.
-- See which of them went to the ATM.
-----------------------------------------------------------
-- 4. Which of these withdrew money on Leggett Street that day?
-- Objective:
-- Use Eugene's clue to filter the suspects.
SELECT people.name
FROM people
JOIN bank_accounts
ON people.id = bank_accounts.person_id
JOIN atm_transactions
ON bank_accounts.account_number = atm_transactions.account_number
WHERE atm_transactions.year = 2024
  AND atm_transactions.month = 7
  AND atm_transactions.day = 28
  AND atm_transactions.atm_location = 'Leggett Street'
  AND atm_transactions.transaction_type = 'withdraw'
  AND people.name
  IN ('Vanessa', 'Bruce', 'Barry', 'Luca', 'Sofia', 'Iman', 'Diana', 'Kelsey');
-- Notes:
-- Remaining: Bruce, Diana, Iman, Luca.
-- See now who made that quick call.
-----------------------------------------------------------
-- 5. Who made calls <60s?
-- Objective:
-- Catch the moment the thief called the accomplice.
SELECT p1.name AS caller_name,
       p2.name AS receiver_name,
       phone_calls.duration
FROM phone_calls
JOIN people AS p1
ON phone_calls.caller = p1.phone_number
JOIN people AS p2
ON phone_calls.receiver = p2.phone_number
WHERE phone_calls.year = 2024
  AND phone_calls.month = 7
  AND phone_calls.day = 28
  AND phone_calls.duration < 60
  AND p1.name
  IN ('Bruce', 'Diana', 'Iman', 'Luca');
-- Notes:
-- Bruce >> Robin (45s)
-- Diana >> Philip (49s)
-- Only these fit. Last piece: who caught the earliest flight?
-----------------------------------------------------------
-- 6. See who caught the first flight the next day
-- Objective:
-- Everything together: flights + passengers + calls >> thief + destination + accomplice.
SELECT p.name,
       a.city AS destination_city,
       p2.name AS accomplice
FROM flights f
JOIN passengers pa
ON f.id = pa.flight_id
JOIN people p
```

```
ON pa.passport_number = p.passport_number
JOIN airports a
ON f.destination_airport_id = a.id
JOIN phone_calls AS pc
ON pc.caller = p.phone_number
JOIN people p2
ON p2.phone_number = pc.receiver
WHERE f.year = 2024
AND f.month = 7
AND f.day = 29
AND pc.duration < 60
AND p.name
IN ('Bruce', 'Diana')
ORDER BY hour, minute
LIMIT 1;
-- Final notes:
-- The thief is Bruce.
-- He escaped on the earlier flight to New York City.
-- The accomplice is Robin, the guy he called.
The THIEF is: Bruce
The city the thief ESCAPED TO: New York City
The ACCOMPLICE is: Robin
```

me50 / users / franciscpime / cs50 / problems / 2025 / x / fiftyville



Week 3

These websites were developed and tested in a local environment. To properly view and run the site as intended, it is necessary to use the five server extension in Visual Studio code, with the project opened from the root folder.

The five server extension allows the website to be served locally, ensuring that all files, links, and resources function correctly. Without these extension, some features of the website may not display or behave as expected.

Trivia

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@500&display=swap"
 rel="stylesheet">
    <link href="./styles.css" rel="stylesheet">
    <title>Trivia!</title>
    <script defer src="./script.js"></script>
</head>
<body>
    <div class="header">
        <h1>Trivia!</h1>
    </div>
    <div class="container">
        <div class="section">
            <h2>Part 1: Multiple Choice </h2>
            <hr>
            <h3>Question 1: What is the capital of France?</h3>
            <label class="label">
                <input class="radio_input" type="radio" name="city" value="Berlin">
                <span class="radio_input-text cool">A) Berlin</span>
            </label>
            <label class="label">
                <input class="radio_input" type="radio" name="city" value="Paris">
                <span class="radio_input-text cool">B) Paris</span>
            </label>
            <label class="label">
                <input class="radio_input" type="radio" name="city" value="Madrid">
                <span class="radio_input-text cool">C) Madrid</span>
            </label>
            <label class="label">
                <input class="radio_input" type="radio" name="city" value="Rome">
                <span class="radio_input-text cool">D) Rome</span>
            </label>
            <p id="feedback1"></p>
```

```
        </div>
        <div class="section">
            <h2>Part 2: Free Response</h2>
            <hr>
            <h3>Who is Donald Trump?</h3>
            <input id="answer2" type="text">
            <button id="check">Check Answer</button>
            <p id="feedback2"></p>
        </div>
    </div>
</body>
</html>
```

CSS

```
body {
    background-color: #fff;
    color: #212529;
    font-size: 1rem;
    font-weight: 400;
    line-height: 1.5;
    margin: 0;
    text-align: left;
}
.container {
    margin-left: auto;
    margin-right: auto;
    padding-left: 15px;
    padding-right: 15px;
}
.header {
    background-color: #477bff;
    color: #fff;
    margin-bottom: 1rem;
    padding: 2rem 1rem;
    text-align: center;
}
.section {
    padding: 0.5rem 2rem 1rem 2rem;
}
.section:hover {
    background-color: #f5f5f5;
    transition: color 2s ease-in-out, background-color 0.15s ease-in-out;
}
h1 {
    font-family: 'Montserrat', sans-serif;
    font-size: 48px;
}
.cool, button, input[type="submit"] {
    background-color: #d9edff;
    border: 1px solid transparent;
    border-radius: 0.25rem;
    font-size: 0.95rem;
    font-weight: 400;
    line-height: 1.5;
    padding: 0.375rem 0.75rem;
    text-align: center;
```

```
    transition: color 0.15s ease-in-out, background-color 0.15s ease-in-out, border-
color 0.15s ease-in-out, box-shadow 0.15s ease-in-out;
    vertical-align: middle;
}
.cool {
    transition: background-color .3s ease-in-out;
}
.label {
    margin-right: 12px;
}
.radio_input {
    position: absolute;
    visibility: hidden;
}
```

JavaScript

```
document.querySelectorAll('input[name="city"]').forEach(radio => {
    radio.addEventListener("change", () => {
        document.querySelectorAll(".radio_input-text").forEach(span => {
            span.style.backgroundColor = "#d9edff";
            span.style.color = "black";
        });
        const selectedSpan = radio.nextElementSibling;
        if (radio.value === "Paris") {
            selectedSpan.style.backgroundColor = "green";
            selectedSpan.style.color = "white";
            document.getElementById("feedback1").textContent = "Correct!";
            document.getElementById("feedback1").style.color = "green";
        } else {
            selectedSpan.style.backgroundColor = "red";
            selectedSpan.style.color = "white";
            document.getElementById("feedback1").textContent = "Incorrect.";
            document.getElementById("feedback1").style.color = "red";
        }
    });
});
document.getElementById("check").addEventListener("click", () => {
    const answer = document.getElementById("answer2").value.trim();
    const feedback2 = document.getElementById("feedback2");
    if (answer === "President") {
        feedback2.textContent = "Correct!";
        feedback2.style.color = "green";
    } else {
        feedback2.textContent = "Incorrect.";
        feedback2.style.color = "red";
    }
});
```

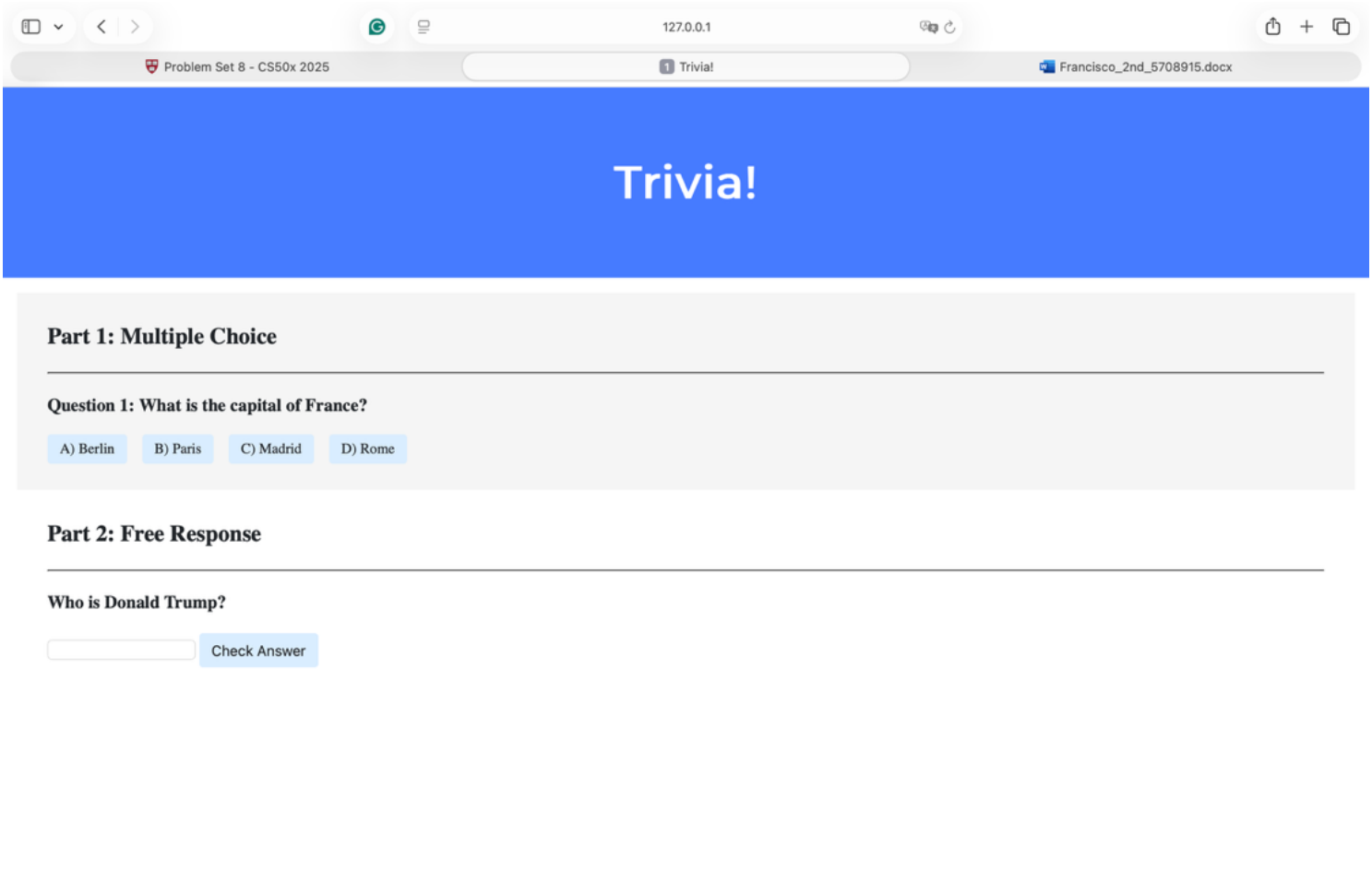


Figure 11 - Website | Default

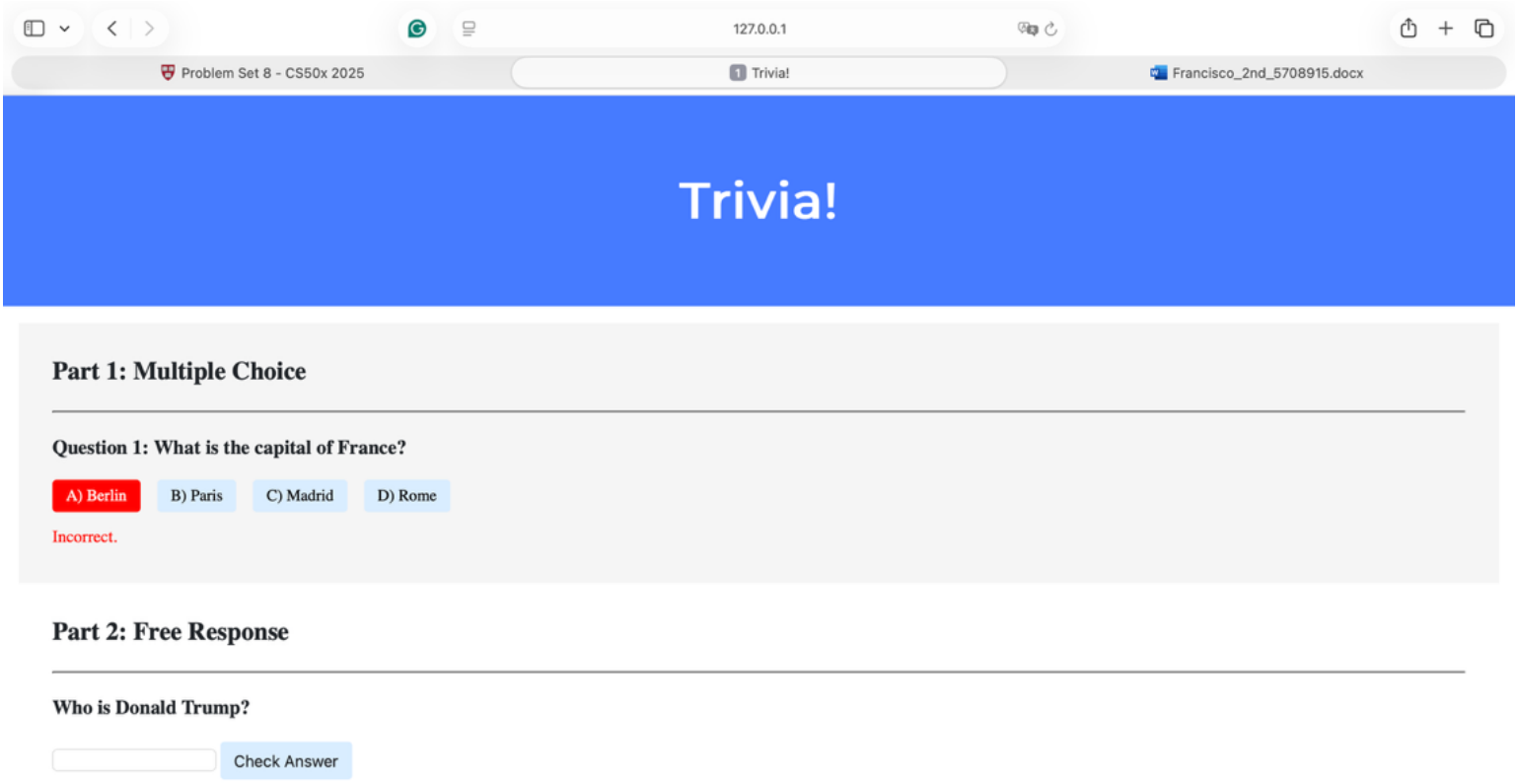


Figure 12 - Website | Exercise1 wrong

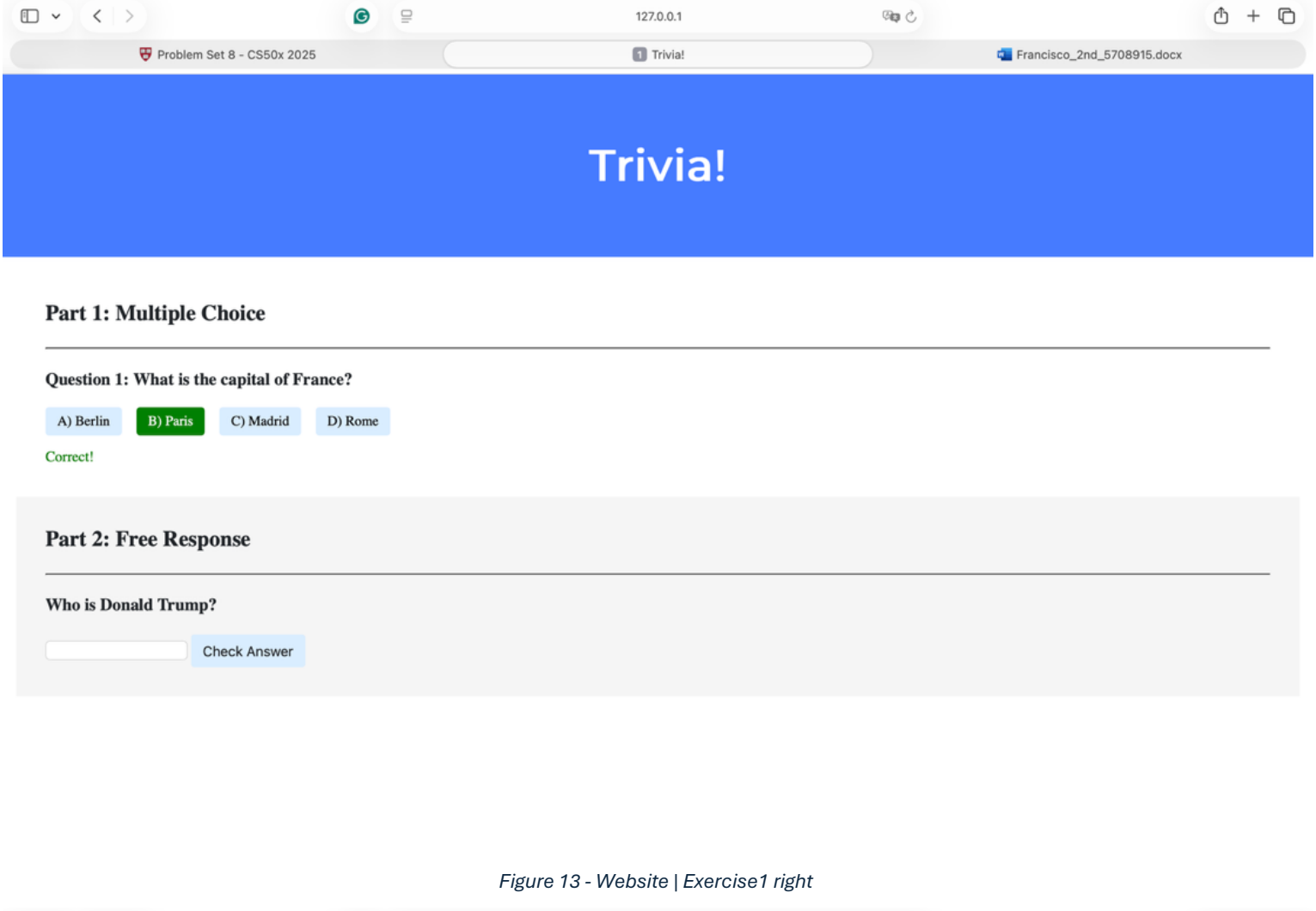


Figure 13 - Website | Exercise1 right

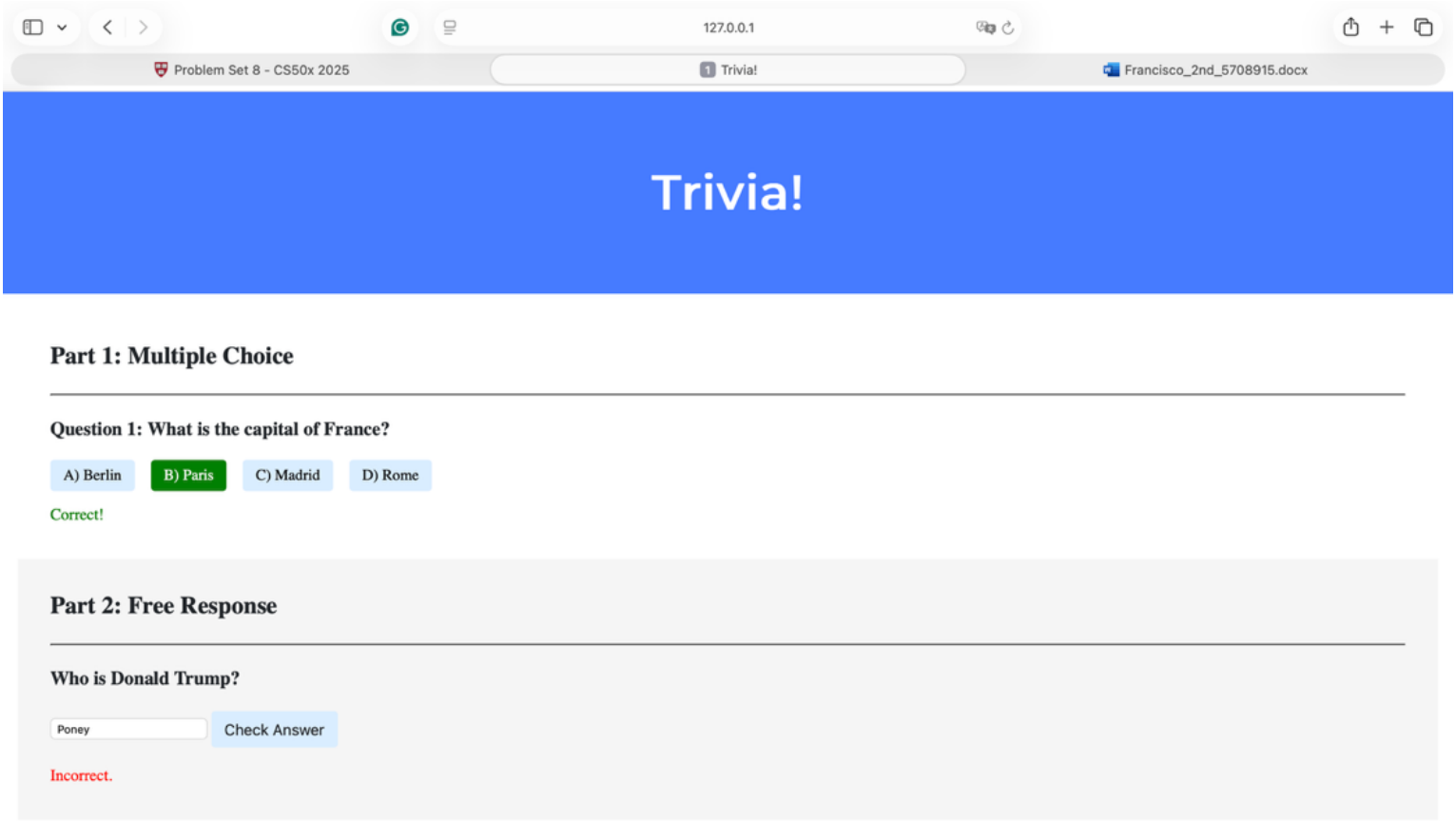


Figure 14 - Website | Exercise2 wrong

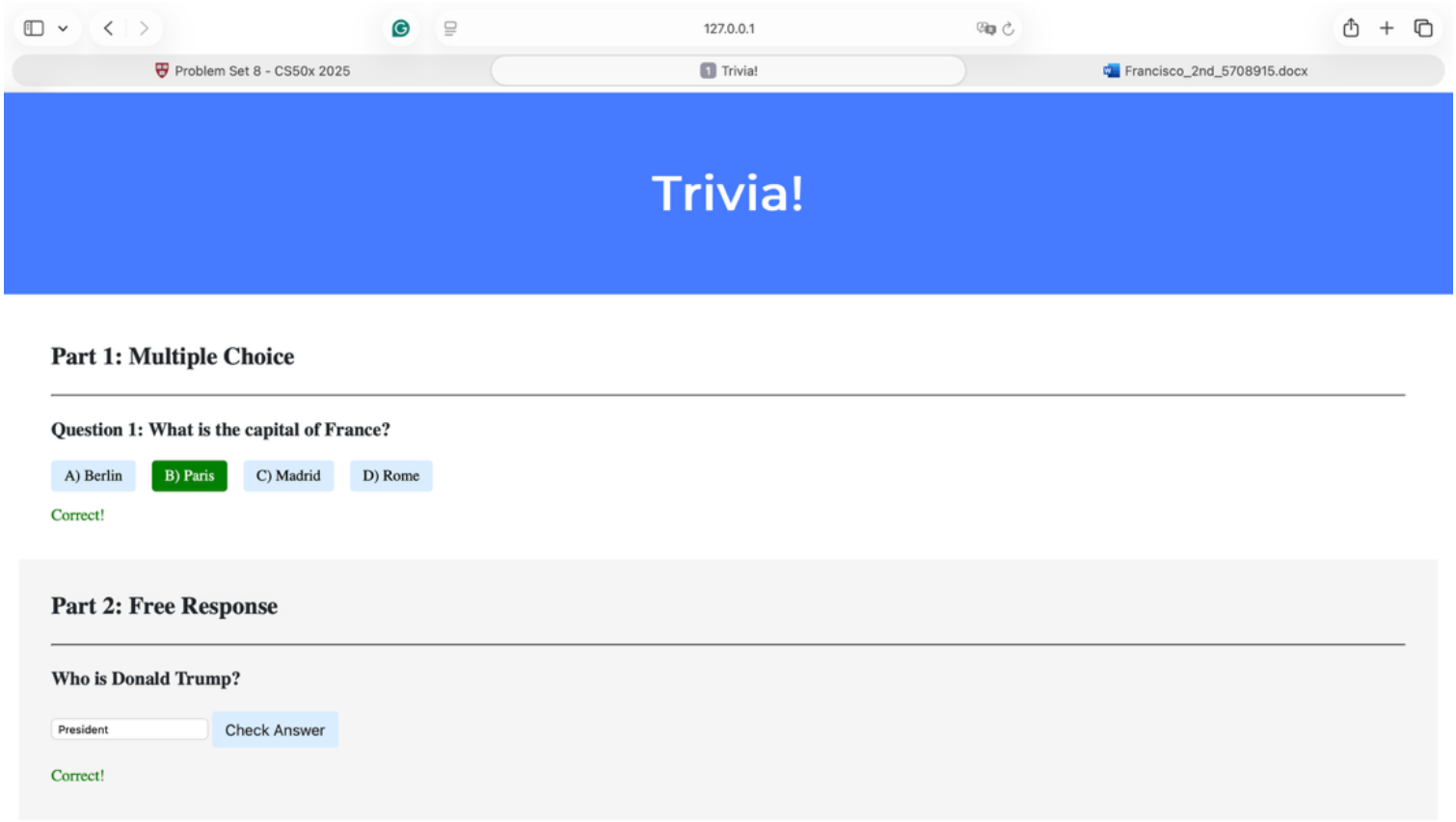
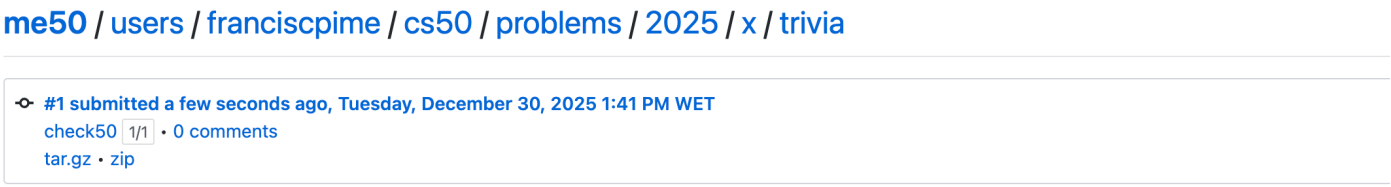


Figure 15 - Website | Exercise2 right



Homepage

Home

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Francisco's Homepage</title>
    <!-- Bootstrap -->
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/boo
tstrap.min.css">
    <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.bundle.min
.js"></script>
    <link href="styles.css" rel="stylesheet">
</head>
<body>
    <!-- navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="navbar-nav mx-auto">
            <a class="nav-item nav-link active" href="index.html">Home</a>
            <a class="nav-item nav-link" href="about.html">About Me</a>
            <a class="nav-item nav-link" href="hobbies.html">Hobbies</a>
            <a class="nav-item nav-link" href="contact.html">Contact</a>
        </div>
    </nav>
    <header class="jumbotron text-center bg-primary text-white">
       <h1 class="display-4 home-title">Welcome to Francisco's Homepage!</h1>
       <p class="lead">Use the navigation bar above to visit the different pages.</p>
    </header>
    <!-- Images -->
    <div class="container mt-5">
        <div class="row text-center">
            <div class="col-md-4 mb-4">
                <img src="images/Francisco1.jpeg"
                     class="img-fluid rounded shadow homepage-img"
                     alt="Francisco Photo 1">
            </div>
            <div class="col-md-4 mb-4">
                <img src="images/Francisco2.jpeg"
                     class="img-fluid rounded shadow homepage-img"
                     alt="Francisco Photo 2">
            </div>
            <div class="col-md-4 mb-4">
                <img src="images/Francisco3.jpeg"
                     class="img-fluid rounded shadow homepage-img"
                     alt="Francisco Photo 3">
            </div>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

About Me

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>About Me</title>
    <!-- Bootstrap -->
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/boo
tstrap.min.css">
    <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.bundle.min
.js"></script>
    <link href="styles.css" rel="stylesheet">
</head>
<body>
    <!-- navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="navbar-nav mx-auto">
            <a class="nav-item nav-link" href="index.html">Home</a>
            <a class="nav-item nav-link active" href="about.html">About Me</a>
            <a class="nav-item nav-link" href="hobbies.html">Hobbies</a>
            <a class="nav-item nav-link" href="contact.html">Contact</a>
        </div>
    </nav>
    <!-- Content -->
    <div class="container mt-5 mb-5">
        <h1 class="mb-4">About Me</h1>
        <!-- Introduction -->
        <div class="row align-items-center mb-5">
            <div class="col-md-6">
                <p>
                    Hi! I'm Francisco, a student passionate about languages, technology, s
ports,
                    and personal growth. This website is part of my CS50 journey, where I
learned
                    the basics of web development and programming.
                </p>
                <p>
                    I enjoy learning new things, taking on challenges, and improving every
 day,
                    whether in sports, university, or coding.
                </p>
            </div>
            <div class="col-md-6 text-center">
                <img src="https://images.unsplash.com/photo-1518770660439-4636190af475"
                     class="img-fluid rounded shadow-lg"
                     alt="Computer">
            </div>
        </div>
        <hr class="my-5">
        <!-- My skills -->
        <h2 class="mb-3">Skills</h2>
        <ul class="ml-3 mb-5">
            <li>Programming (C, C++, Python)</li>
            <li>Languages (Portuguese, English, Spanish, French, Japanese)</li>
            <li>Roller Hockey</li>
```

```
            <li>Strong motivation and work ethic</li>
        </ul>
        <hr class="my-5">
        <!-- My journey -->
        <h2 class="mb-3">My Journey</h2>
        <p>Here's a quick timeline of some important steps in my learning path:</p>
        <ul class="ml-3">
            <li><strong>2012:</strong> Started roller hockey – still going today!</li>
            <li><strong>2024:</strong> Began learning programming.</li>
            <li><strong>2025:</strong> Finished high school.</li>
            <li><strong>2025:</strong> Joined CS50 and started building real projects.</li
>
        </ul>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

Hobbies

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Hobbies</title>
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/boo
tstrap.min.css">
    <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.bundle.min
.js"></script>
    <link href="styles.css" rel="stylesheet">
</head>
<body>
    <!-- navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="navbar-nav mx-auto">
            <a class="nav-item nav-link" href="index.html">Home</a>
            <a class="nav-item nav-link" href="about.html">About Me</a>
            <a class="nav-item nav-link active" href="hobbies.html">Hobbies</a>
            <a class="nav-item nav-link" href="contact.html">Contact</a>
        </div>
    </nav>
    <div class="container mt-4">
        <h1>My Hobbies</h1>
        <p class="text-center">These are the activities that I enjoy the most.</p>
        <div class="row mt-4">
            <!-- hobby 1 -->
            <div class="col-md-6">
                <div class="card hobby-card">
                    <img src="images/hockey.png" class="card-img-top" alt="Roller Hockey">
                    <div class="card-body">
                        <h5 class="card-title">Hockey</h5>
                        <p class="card-text">
                     I've played roller hockey for 13 years – it's a huge part of my life.
                     The discipline, teamwork and speed of the sport inspire me every day.
                        </p>
```

```
                    </div>
                </div>
            </div>
            <!-- hobby 2 -->
            <div class="col-md-6">
                <div class="card hobby-card">
                    <img src="images/gym.jpeg" class="card-img-top" alt="Gymnastics">
                    <div class="card-body">
                        <h5 class="card-title">Gym</h5>
                        <p class="card-text">
                          Gym helps me stay healthy and strong. And it's thanks to it that
                          I've learned discipline and consistency.
                        </p>
                    </div>
                </div>
            </div>
        </div>
        <div class="row mt-4">
            <!-- hobby 3 -->
            <div class="col-md-6">
                <div class="card hobby-card">
                    <img src="images/programming.jpeg" class="card-img-top" alt="Coding">
                    <div class="card-body">
                        <h5 class="card-title">Programming</h5>
                        <p class="card-text">
           I love coding because it lets me solve problems and create things from scratch.
           This website is one of the results!
                        </p>
                    </div>
                </div>
            </div>
            <!-- hobby 4 -->
            <div class="col-md-6">
                <div class="card hobby-card">
                    <img src="images/languages.png"
                        class="card-img-top" alt="Languages">
                    <div class="card-body">
                        <h5 class="card-title">Learning Languages</h5>
                        <p class="card-text">
           I’m learning French and Japanese.
           Knowing new languages helps me connect with people and understand new cultures.
                        </p>
                    </div>
                </div>
            </div>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

Contact

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Contact Me</title>
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/boo
tstrap.min.css">
    <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.bundle.min
.js"></script>
    <link href="styles.css" rel="stylesheet">
</head>
<body>
    <!-- navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="navbar-nav mx-auto">
            <a class="nav-item nav-link" href="index.html">Home</a>
            <a class="nav-item nav-link" href="about.html">About Me</a>
            <a class="nav-item nav-link" href="hobbies.html">Hobbies</a>
            <a class="nav-item nav-link active" href="contact.html">Contact</a>
        </div>
    </nav>
    <div class="container mt-4">
        <h1>Contact Me</h1>
        <p>Send me a quick message:</p>
        <form id="contact-form">
            <div class="form-group">
                <label>Your name:</label>
                <input id="name-field" type="text" class="form-control">
            </div>
            <div class="form-group">
                <label>Your email:</label>
                <input id="email-field" type="email" class="form-control">
            </div>
            <div class="form-group">
                <label>Your message:</label>
                <textarea id="message-field" class="form-control" rows="4"></textarea>
            </div>
            <button type="button" id="say-hello" class="btn btn-primary">Send</button>
        </form>
        <div id="response" class="mt-3"></div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

CSS

```css
body {
    background-color: #f8f9fa;
}
p, li {
    font-size: 18px;
}
/* Page titles */
h1 {
    text-align: center;
    margin-bottom: 20px;
}
/* Homepage title */
.home-title {
    color: white !important;
}
/* Titles for other pages */
.dark-title {
    color: #222 !important;
}
.homepage-img {
    width: 100%;
    height: 350px;
    object-fit: cover;
    border-radius: 10px;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.homepage-img:hover {
    transform: scale(1.03);
    box-shadow: 0 6px 15px rgba(0, 0, 0, 0.3);
}
.card {
    border: 2px solid #007bff;
    border-radius: 10px;
    overflow: hidden;
    height: 100%;
}
.card-img-top {
    width: 100%;
    height: 230px;
    object-fit: cover;
}
#response {
    font-weight: bold;
    margin-top: 20px;
}
img {
    max-width: 100%;
}
```

Script

```
document.addEventListener("DOMContentLoaded", function() {
    const button = document.getElementById("say-hello");
    if (button) {
        // Assign a function to run when the button is clicked.
        button.onclick = function() {
          // Retrieve the form fields and the area where the response message will appear.
            const nameField = document.getElementById("name-field");
            const emailField = document.getElementById("email-field");
            const messageField = document.getElementById("message-field");
            const response = document.getElementById("response");
            // These start empty and are filled only if the fields exist.
            let name = "";
            let email = "";
            let message = "";
            // Extract name value if the field exists.
            if (nameField) {
                name = nameField.value.trim();
            }
            // Extract email value if the field exists.
            if (emailField) {
                email = emailField.value.trim();
            }
            // Extract message value if the field exists.
            if (messageField) {
                message = messageField.value.trim();
            }
            // Check if the user typed a name.
            if (name.length === 0) {
                response.innerHTML =
                    '<div class="alert alert-danger">Please enter your name.</div>';
                return;
            }
            // Accept only emails that end with '@gmail.com'.
            if (email.length > 0 && !email.endsWith("@gmail.com")) {
                response.innerHTML =
                    '<div class="alert alert-
danger">Only gmail addresses are accepted.</div>';
                return;
            }
            // Check if a message was written but is too short.
            if (message.length > 0 && message.length < 5) {
                response.innerHTML =
                    '<div class="alert alert-danger">Your message is too short.</div>';
                return;
            }
            // If all fields are valid, show a Bootstrap success alert.
            response.innerHTML =
                `<div class="alert alert-
success">Hello, ${name}! Your message has been received.</div>`;
        };
    }
});
```

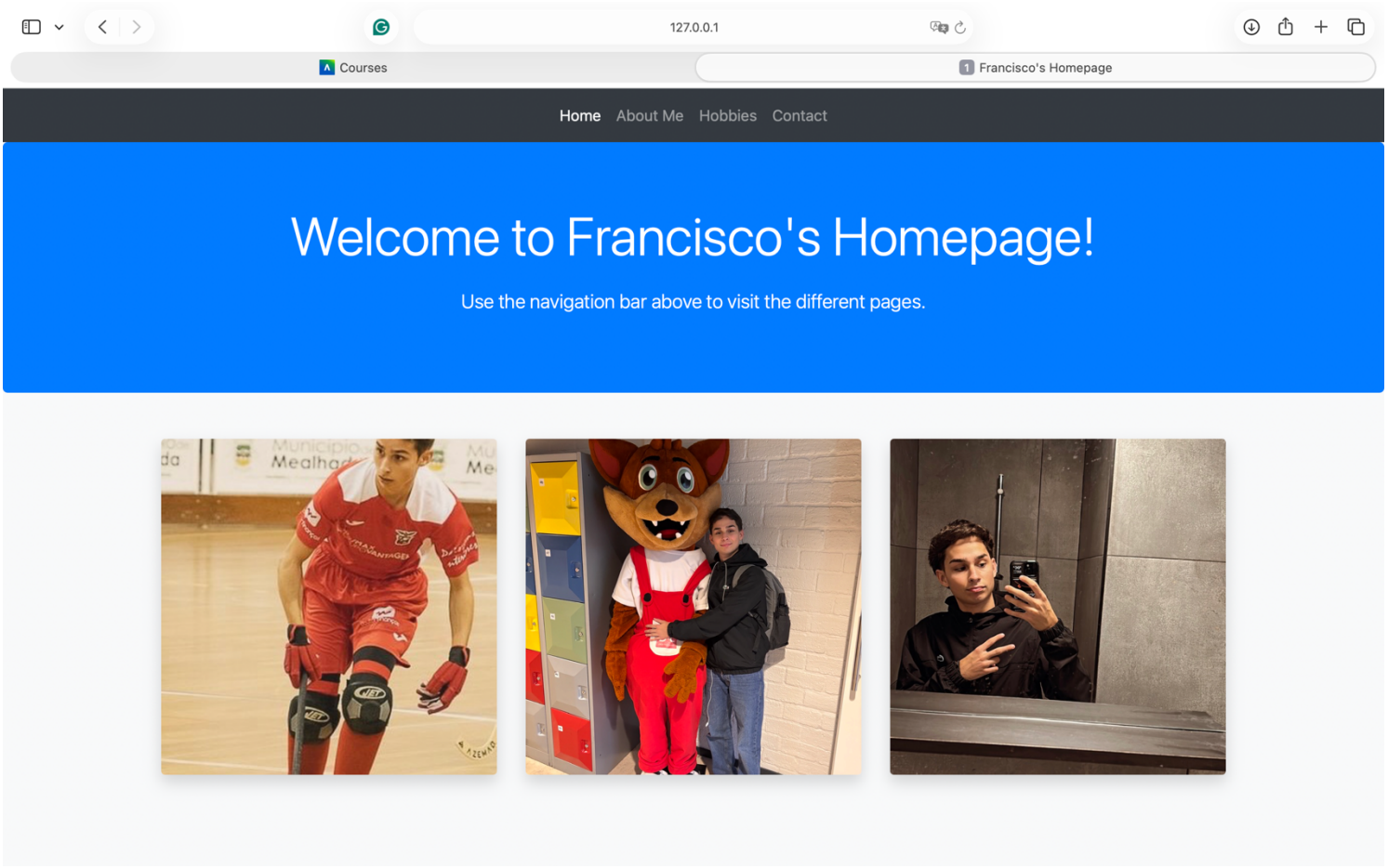


Figure 16 - Websites | Homepage

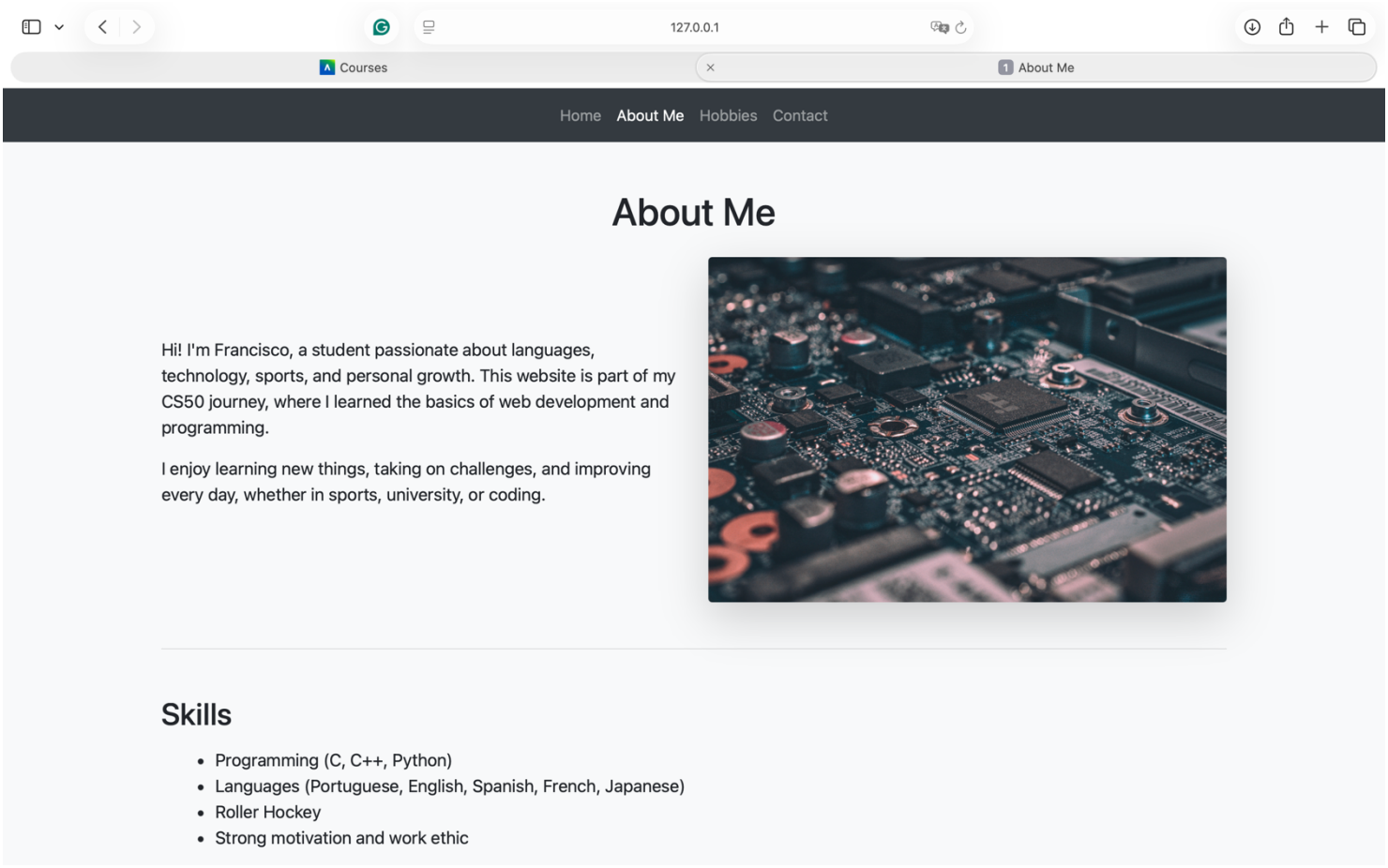
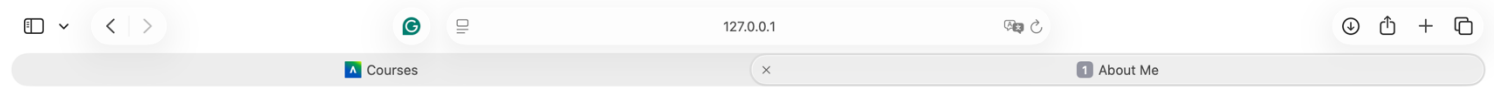
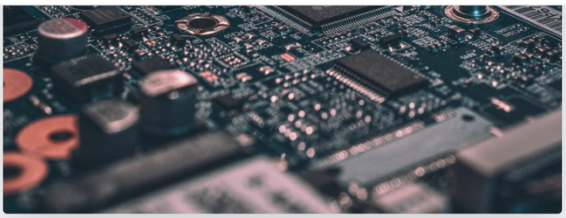


Figure 17 - Website | Aboute Me 1



programming.

I enjoy learning new things, taking on challenges, and improving every day, whether in sports, university, or coding.



Skills

- Programming (C, C++, Python)
- Languages (Portuguese, English, Spanish, French, Japanese)
- Roller Hockey
- Strong motivation and work ethic

My Journey

Here's a quick timeline of some important steps in my learning path:

- **2012:** Started roller hockey — still going today!
- **2024:** Began learning programming.
- **2025:** Finished high school.
- **2025:** Joined CS50 and started building real projects.

Figure 18 - Website | About Me 2

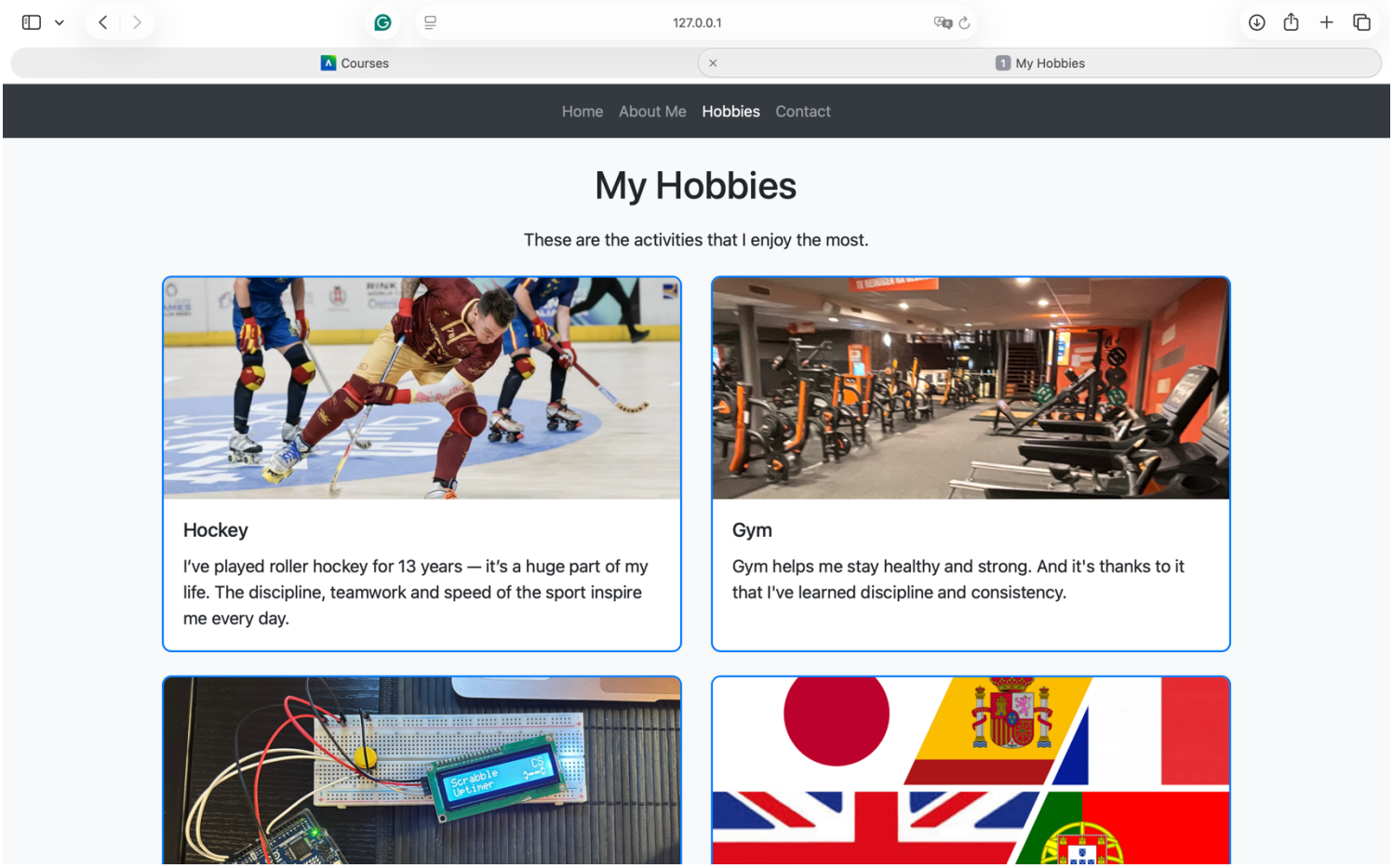


Figure 19 - Website | Hobbies 1

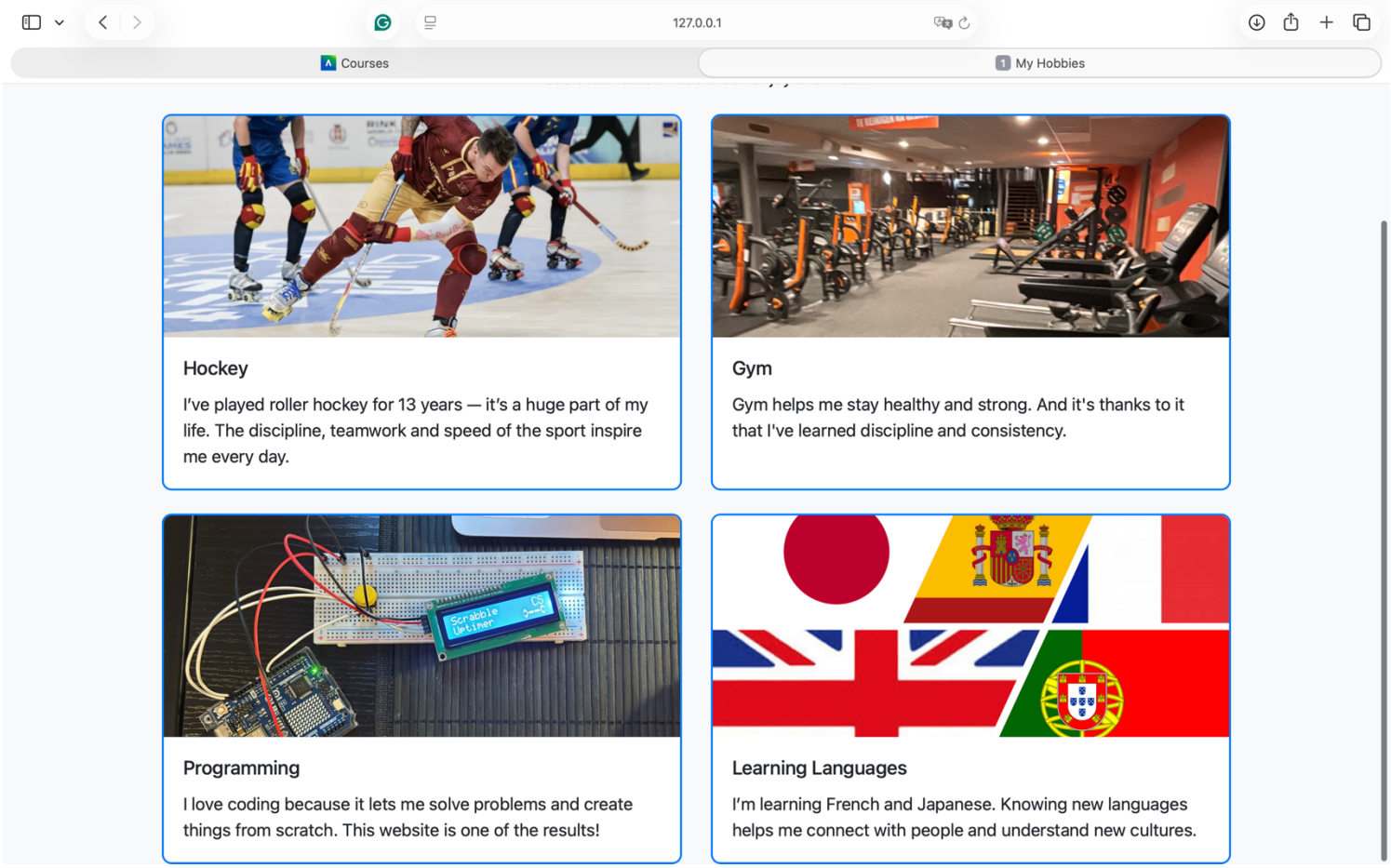


Figure 20 - Website | Hobbies 2

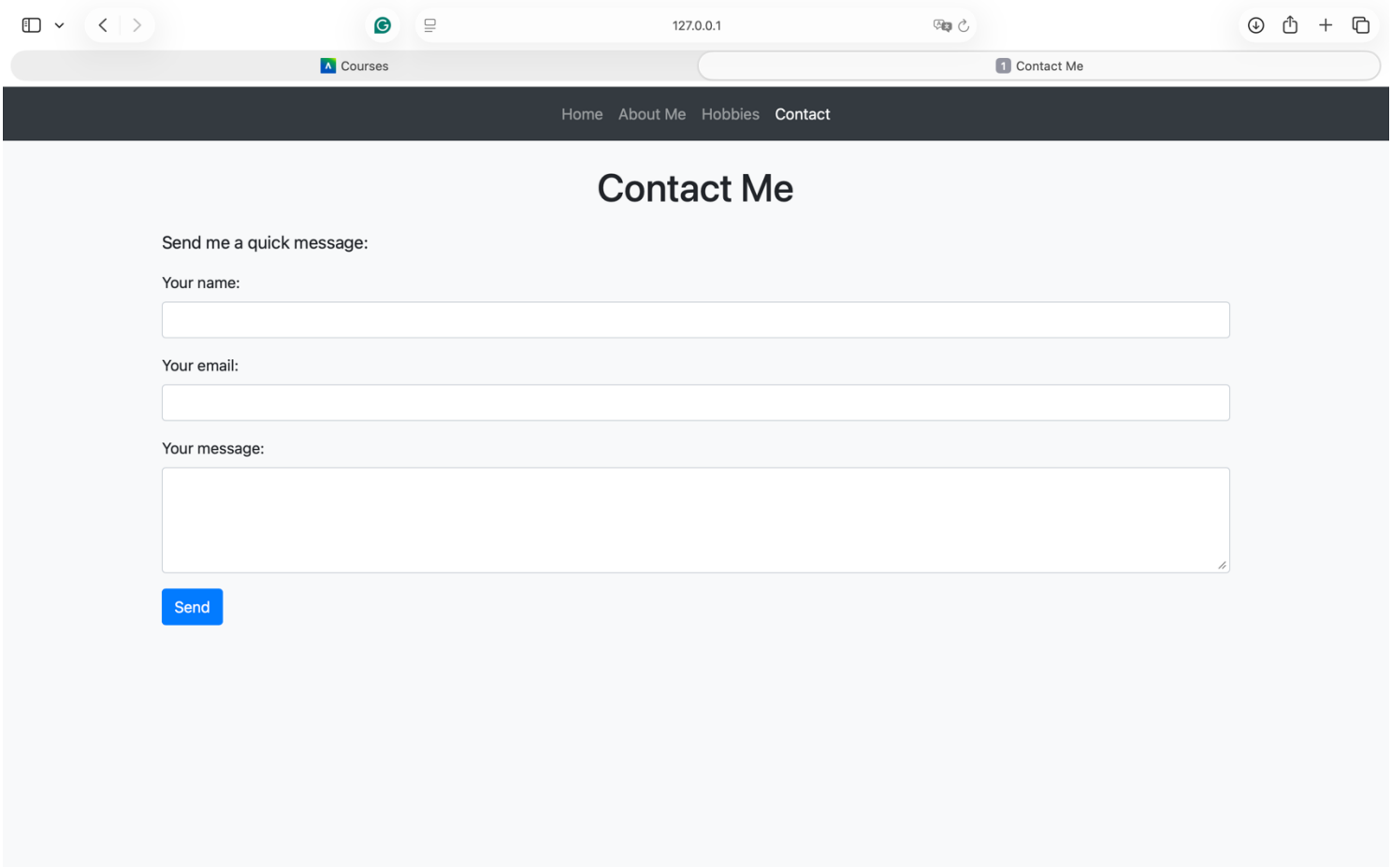


Figure 21 - Website | Contact

me50 / users / franciscpime / cs50 / problems / 2025 / x / homepage

#1 submitted a few seconds ago, Tuesday, December 30, 2025 1:45 PM WET
check50 1/1 • 0 comments
tar.gz • zip

Week 4

Birthdays

HTML

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@500&display=s
wap" rel="stylesheet">
        <link href="/static/styles.css" rel="stylesheet">
        <title>Birthdays</title>
    </head>
    <body>
        <div class="header">
            <h1>Birthdays</h1>
        </div>
        <div class="container">
            <div class="section">
                <h2>Add a Birthday</h2>
                <!-- Form to submit new birthdays -->
                <form action="/" method="post">
          <input name="name" type="text" placeholder="Name" required>
          <input name="month" type="number" placeholder="Month" min="1" max="12" required>
          <input name="day" type="number" placeholder="Day" min="1" max="31" required>
          <input type="submit" value="Add Birthday">
                </form>
            </div>
            <div class="section">
                <h2>All Birthdays</h2>
                <table>
                    <thead>
                        <tr>
                            <th>Name</th>
                            <th>Birthday</th>
                        </tr>
                    </thead>
                    <tbody>
                        <!-- Loop through birthdays from the database -->
                        {% for birthday in birthdays %}
                        <tr>
                            <td>{{ birthday.name }}</td>
                            <td>{{ birthday.month }}/{{ birthday.day }}</td>
                        </tr>
                        {% endfor %}
                    </tbody>
                </table>
            </div>
        </div>
    </body>
</html>
```

CSS

```
body {
    background-color: #fff;
    color: #212529;
    font-size: 1rem;
    font-weight: 400;
    line-height: 1.5;
    margin: 0;
    text-align: left;
}
.container {
    margin-left: auto;
    margin-right: auto;
    padding-left: 15px;
    padding-right: 15px;
    text-align: center;
    width: 90%;
}
.header {
    background-color: #477bff;
    color: #fff;
    margin-bottom: 2rem;
    padding: 2rem 1rem;
    text-align: center;
}
.section {
    padding-bottom: 1rem;
    padding-left: 2rem;
    padding-right: 2rem;
    padding-top: 0.5rem;
}
.section:hover {
    background-color: #f5f5f5;
    transition: color 2s ease-in-out, background-color 0.15s ease-in-out;
}
h1 {
    font-family: 'Montserrat', sans-serif;
    font-size: 48px;
}
button, input[type="submit"] {
    background-color: #d9edff;
    border: 1px solid transparent;
    border-radius: 0.25rem;
    font-size: 0.95rem;
    font-weight: 400;
    line-height: 1.5;
    padding: 0.375rem 0.75rem;
    text-align: center;
    transition: color 0.15s ease-in-out, background-color 0.15s ease-in-out, border-
color 0.15s ease-in-out, box-shadow 0.15s ease-in-out;
    vertical-align: middle;
}
input[type="text"], input[type="number"] {
    line-height: 1.8;
    width: 25%;
}
input[type="text"]:hover, input[type="number"]:hover {
```

```
    background-color: #f5f5f5;
    transition: color 2s ease-in-out, background-color 0.15s ease-in-out;
}
table {
    background-color: transparent;
    margin-bottom: 1rem;
    width: 100%;
}
table th, table td {
    padding: 0.75rem;
    vertical-align: middle;
}
tbody tr:nth-of-type(odd) {
    background-color: rgb(179, 208, 255, 0.3)
}
```

APP

```
import os
from cs50 import SQL
from flask import Flask, flash, jsonify, redirect, render_template, request, session
# Configure application
app = Flask(__name__)
# Ensure templates are auto-reloaded
app.config["TEMPLATES_AUTO_RELOAD"] = True
# Configure CS50 Library to use SQLite database
db = SQL("sqlite:///birthdays.db")
@app.after_request
def after_request(response):
    """Ensure responses aren't cached"""
    response.headers["Cache-Control"] = "no-cache, no-store, must-revalidate"
    response.headers["Expires"] = 0
    response.headers["Pragma"] = "no-cache"
    return response
@app.route("/", methods=["GET", "POST"])
def index():
    if request.method == "POST":
        # Get data from the form
        name = request.form.get("name")
        month = request.form.get("month")
        day = request.form.get("day")
        # Validate basic input
        if not name or not month or not day:
            return redirect("/")
        try:
            month = int(month)
            day = int(day)
        except ValueError:
            return redirect("/")
        if month < 1 or month > 12:
            return redirect("/")
        if day < 1 or day > 31:
            return redirect("/")
        # Insert birthday into database
        db.execute(
            "INSERT INTO birthdays (name, month, day) VALUES (?, ?, ?)",
            name, month, day
```

```
        )
        return redirect("/")
    else:
        # Query for all birthdays
        birthdays = db.execute("SELECT * FROM birthdays")
        # Render the page WITH the birthdays data
        return render_template("index.html", birthdays=birthdays)
```

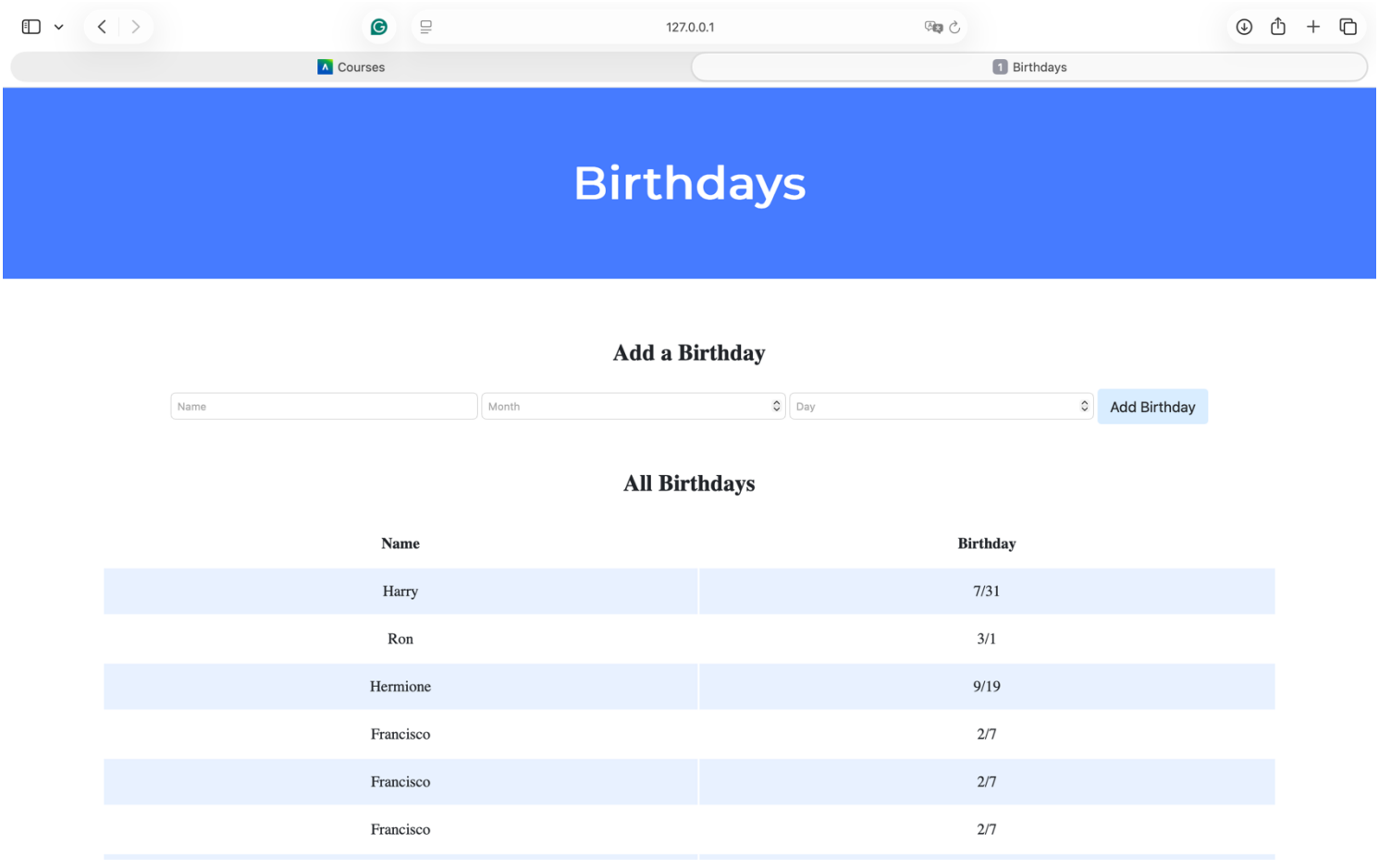


Figure 22 - Website | Birthdays

me50 / users / franciscpime / cs50 / problems / 2025 / x / birthdays

#1 submitted a few seconds ago, Tuesday, December 30, 2025 2:37 PM WET
check50 1/1 • style50 0.98 • 0 comments
tar.gz • zip

Finance

Apology

```
{% extends "layout.html" %}
{% block title %}
    Apology
{% endblock %}
{% block main %}
    <!-- https://memegen.link/ -->
    <!-- https://knowyourmeme.com/memes/grumpy-cat -->
    <img alt="{{ top }}" class="border img-
fluid" src="https://api.memegen.link/images/custom/{{ top | urlencode }}/{{ bottom | urlen
code }}.jpg?background=https://i.imgur.com/CsCgN7Ll.png&width=400" title="{{ top }}">
{% endblock %}
```

Buy

```
{% extends "layout.html" %}
{% block main %}
<form action="/buy" method="post">
    <div class="mb-3">
        <input class="form-control"
               name="symbol"
               placeholder="Symbol"
               type="text">
    </div>
    <div class="mb-3">
        <input class="form-control"
               name="shares"
               placeholder="Shares"
               type="number" min="1">
    </div>
    <button class="btn btn-primary" type="submit">Buy</button>
</form>
{% endblock %}
```

Change Password

```
{% extends "layout.html" %}
{% block main %}
<form action="/password" method="post">
    <div class="mb-3">
        <input class="form-control"
               name="old_password"
               placeholder="Current Password"
               type="password">
    </div>
    <div class="mb-3">
        <input class="form-control"
               name="new_password"
               placeholder="New Password"
               type="password">
    </div>
```

```
    <div class="mb-3">
        <input class="form-control"
               name="confirmation"
               placeholder="Confirm New Password"
               type="password">
    </div>
    <button class="btn btn-primary" type="submit">
        Change Password
    </button>
</form>
{% endblock %}
```

History

```
{% extends "layout.html" %}
{% block main %}
<table class="table">
    <thead>
        <tr>
            <th>Type</th>
            <th>Symbol</th>
            <th>Shares</th>
            <th>Price</th>
            <th>Date</th>
        </tr>
    </thead>
    <tbody>
        {% for row in history %}
        <tr>
            <td>{{ row.type }}</td>
            <td>{{ row.symbol }}</td>
            <td>{{ row.shares }}</td>
            <td>{{ row.price | usd }}</td>
            <td>{{ row.timestamp }}</td>
        </tr>
        {% endfor %}
    </tbody>
</table>
{% endblock %}
```

Index

```
{% extends "layout.html" %}
{% block main %}
<table class="table">
    <thead>
        <tr>
            <th>Symbol</th>
            <th>Shares</th>
            <th>Price</th>
            <th>Total</th>
        </tr>
    </thead>
    <tbody>
        {% for stock in holdings %}
        <tr>
```

```
            <td>{{ stock.symbol }}</td>
            <td>{{ stock.shares }}</td>
            <td>{{ stock.price | usd }}</td>
            <td>{{ stock.total | usd }}</td>
        </tr>
        {% endfor %}
    </tbody>
    <tfoot>
        <tr>
            <td colspan="3">Cash</td>
            <td>{{ cash | usd }}</td>
        </tr>
        <tr>
            <td colspan="3"><strong>Grand Total</strong></td>
            <td><strong>{{ total | usd }}</strong></td>
        </tr>
    </tfoot>
</table>
{% endblock %}
```

Layout

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="initial-scale=1, width=device-width">
        <!-- http://getbootstrap.com/docs/5.3/ -->
        <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.cs
s" rel="stylesheet" integrity="sha384-
QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhjY6hW+ALEwIH" crossorigin="anonymous">
        <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle
.min.js" integrity="sha384-
YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz" crossorigin="anonymous">
</script>
        <!-- https://favicon.io/emoji-favicons/money-bag/ -->
        <link href="/static/favicon.ico" rel="icon">
        <link href="/static/styles.css" rel="stylesheet">
        <title>C$50 Finance: {% block title %}{% endblock %}</title>
    </head>
    <body>
        <nav class="bg-light border navbar navbar-expand-md navbar-light">
            <div class="container-fluid">
                <a class="navbar-
brand" href="/"><span class="blue">C</span><span class="red">$</span><span class="yellow">
5</span><span class="green">0</span> <span class="red">Finance</span></a>
                <button aria-controls="navbar" aria-expanded="false" aria-
label="Toggle navigation" class="navbar-toggler" data-bs-target="#navbar" data-bs-
toggle="collapse" type="button">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="collapse navbar-collapse" id="navbar">
                    {% if session["user_id"] %}
                        <ul class="navbar-nav me-auto mt-2">
                     <li class="nav-item"><a class="nav-link" href="/quote">Quote</a></li>
                     <li class="nav-item"><a class="nav-link" href="/buy">Buy</a></li>
                     <li class="nav-item"><a class="nav-link" href="/sell">Sell</a></li>
```

```
                <li class="nav-item"><a class="nav-link" href="/history">History</a></li>
                        <li class="nav-item"><a class="nav-
link" href="/password">Change Password</a></li>
                    </ul>
                    <ul class="navbar-nav ms-auto mt-2">
                 <li class="nav-item"><a class="nav-link" href="/logout">Log Out</a></li>
                    </ul>
                {% else %}
                    <ul class="navbar-nav ms-auto mt-2">
             <li class="nav-item"><a class="nav-link" href="/register">Register</a></li>
                <li class="nav-item"><a class="nav-link" href="/login">Log In</a></li>
                    </ul>
                {% endif %}
            </div>
        </div>
    </nav>
    {% if get_flashed_messages() %}
        <header>
            <div class="alert alert-primary mb-0 text-center" role="alert">
                {{ get_flashed_messages() | join(" ") }}
            </div>
        </header>
    {% endif %}
    <main class="container py-5 text-center">
        {% block main %}{% endblock %}
    </main>
    <footer class="mb-5">
        <form action="https://validator.w3.org/check" class="text-
center" enctype="multipart/form-data" method="post" target="_blank">
            <input name="doctype" type="hidden" value="HTML5">
            <input name="fragment" type="hidden">
            <input alt="Validate" src="/static/I_heart_validator.png" type="image">
<!-- https://validator.w3.org/ -->
        </form>
        <script>
            document.addEventListener('DOMContentLoaded', function() {
                // Adapted from https://stackoverflow.com/a/10162353
                const html = '<!DOCTYPE ' +
                document.doctype.name +
    (document.doctype.publicId ? ' PUBLIC "' + document.doctype.publicId + '"' : '') +
    (!document.doctype.publicId && document.doctype.systemId ? ' SYSTEM' : '') +
    (document.doctype.systemId ? ' "' + document.doctype.systemId + '"' : '') +
                '>\n' + document.documentElement.outerHTML;
                document.querySelector('form[action="https://validator.w3.org/check"]
> input[name="fragment"]').value = html;
            });
        </script>
    </footer>
</body>
</html>
```

Login

```
{% extends "layout.html" %}
{% block title %}
    Log In
{% endblock %}
```

```
{% block main %}
    <form action="/login" method="post">
        <div class="mb-3">
            <input autocomplete="off" autofocus class="form-control mx-auto w-
auto" name="username" placeholder="Username" type="text">
        </div>
        <div class="mb-3">
            <input class="form-control mx-auto w-
auto" name="password" placeholder="Password" type="password">
        </div>
        <button class="btn btn-primary" type="submit">Log In</button>
    </form>
{% endblock %}
```

Quote

```
{% extends "layout.html" %}
{% block main %}
<form action="/quote" method="post">
    <div class="mb-3">
        <input autocomplete="off" autofocus class="form-control"
               name="symbol" placeholder="Symbol (e.g. AAPL)" type="text">
    </div>
    <button class="btn btn-primary" type="submit">Get Quote</button>
</form>
{% endblock %}
```

Quoted

```
{% extends "layout.html" %}
{% block main %}
<h2>Quote</h2>
<ul>
    <li><strong>Name:</strong> {{ stock.name }}</li>
    <li><strong>Symbol:</strong> {{ stock.symbol }}</li>
    <li><strong>Price:</strong> {{ stock.price | usd }}</li>
</ul>
<a href="/quote">Look up another stock</a>
{% endblock %}
```

Register

```
{% extends "layout.html" %}
{% block main %}
<form action="/register" method="post">
    <div class="mb-3">
        <input autocomplete="off" autofocus class="form-control"
               name="username" placeholder="Username" type="text">
    </div>
    <div class="mb-3">
        <input class="form-control"
               name="password" placeholder="Password" type="password">
    </div>
    <div class="mb-3">
        <input class="form-control"
               name="confirmation" placeholder="Confirm Password" type="password">
```

```
    </div>
    <button class="btn btn-primary" type="submit">Register</button>
</form>
{% endblock %}
```

Sell

```
{% extends "layout.html" %}
{% block main %}
<form action="/sell" method="post">
    <div class="mb-3">
        <select class="form-select" name="symbol">
            <option disabled selected>Symbol</option>
            {% for symbol in symbols %}
                <option value="{{ symbol }}">{{ symbol }}</option>
            {% endfor %}
        </select>
    </div>
    <div class="mb-3">
        <input class="form-control"
               name="shares"
               placeholder="Shares"
               type="number" min="1">
    </div>
    <button class="btn btn-primary" type="submit">Sell</button>
</form>
{% endblock %}
```

CSS

```
/* Size for brand */
nav .navbar-brand
{
    font-size: xx-large;
}
/* Colors for brand */
nav .navbar-brand .blue
{
    color: #537fbe;
}
nav .navbar-brand .red
{
    color: #ea433b;
}
nav .navbar-brand .yellow
{
    color: #f5b82e;
}
nav .navbar-brand .green
{
    color: #2e944b;
}
```

APP

```
import os
# CS50 SQL library for database interaction
from cs50 import SQL
# Flask core utilities
from flask import Flask, flash, redirect, render_template, request, session
# Server-side session management
from flask_session import Session
# Password hashing utilities
from werkzeug.security import check_password_hash, generate_password_hash
# Helper functions provided by CS50 Finance
from helpers import apology, login_required, lookup, usd
# Configure Flask application
app = Flask(__name__)
# Register custom Jinja filter to format values as USD
app.jinja_env.filters["usd"] = usd
# Configure session to use filesystem (instead of signed cookies)
app.config["SESSION_PERMANENT"] = False
app.config["SESSION_TYPE"] = "filesystem"
Session(app)
# Configure CS50 Library to use SQLite database
db = SQL("sqlite:///finance.db")
@app.after_request
def after_request(response):
    """Ensure responses aren't cached"""
    response.headers["Cache-Control"] = "no-cache, no-store, must-revalidate"
    response.headers["Expires"] = 0
    response.headers["Pragma"] = "no-cache"
    return response
@app.route("/")
@login_required
def index():
    """Show portfolio of stocks"""
    # Get user ID from session
    user_id = session["user_id"]
    # Get current holdings
    rows = db.execute(
        """
        SELECT symbol, SUM(shares) AS shares
        FROM transactions
        WHERE user_id = ?
        GROUP BY symbol
        HAVING shares > 0
        """,
        user_id
    )
    holdings = []
    total_stocks_value = 0
    # Calculate total value for each holding
    for row in rows:
        stock = lookup(row["symbol"])
        price = stock["price"]
        total = price * row["shares"]
        holdings.append({
            "symbol": row["symbol"],
            "shares": row["shares"],
            "price": price,
```

```
            "total": total
        })
        total_stocks_value += total
    # Get user's cash
    cash = db.execute(
        "SELECT cash FROM users WHERE id = ?",
        user_id
    )[0]["cash"]
    # Calculate grand total
    grand_total = total_stocks_value + cash
    return render_template(
        "index.html",
        holdings=holdings,
        cash=cash,
        total=grand_total
    )
@app.route("/buy", methods=["GET", "POST"])
@login_required
def buy():
    """Buy shares of stock"""
    if request.method == "POST":
        # Get form data
        symbol = request.form.get("symbol")
        shares = request.form.get("shares")
        # Validate input
        if not symbol:
            return apology("must provide symbol")
        stock = lookup(symbol)
        if stock is None:
            return apology("invalid symbol")
        # Validate shares
        if not shares:
            return apology("must provide shares")
        try:
            shares = int(shares)
        except ValueError:
            return apology("shares must be a positive integer")
        if shares <= 0:
            return apology("shares must be a positive integer")
        price = stock["price"]
        total_cost = shares * price
        user_id = session["user_id"]
        # Get user's cash
        rows = db.execute(
            "SELECT cash FROM users WHERE id = ?", user_id)
        cash = rows[0]["cash"]
        # Check if user can afford the purchase
        if cash < total_cost:
            return apology("can't afford")
        # Update user's cash
        db.execute(
            "UPDATE users SET cash = cash - ? WHERE id = ?",
            total_cost, user_id
        )
        # Record transaction
        db.execute(
            """INSERT INTO transactions (user_id, symbol, shares, price, type)
```

```
                VALUES (?, ?, ?, ?, 'BUY')""",
            user_id, stock["symbol"], shares, price
        )
        return redirect("/")
    else:
        return render_template("buy.html")
@app.route("/history")
@login_required
def history():
    """Show history of transactions"""
    user_id = session["user_id"]
    # Get all transactions for user
    rows = db.execute(
        """
        SELECT symbol, shares, price, type, timestamp
        FROM transactions
        WHERE user_id = ?
        ORDER BY timestamp
        """,
        user_id
    )
    return render_template("history.html", history=rows)
@app.route("/login", methods=["GET", "POST"])
def login():
    """Log user in"""
    # Forget any user_id
    session.clear()
    # User reached route via POST (as by submitting a form via POST)
    if request.method == "POST":
        # Ensure username was submitted
        if not request.form.get("username"):
            return apology("must provide username", 403)
        # Ensure password was submitted
        elif not request.form.get("password"):
            return apology("must provide password", 403)
        # Query database for username
        rows = db.execute(
            "SELECT * FROM users WHERE username = ?", request.form.get("username")
        )
        # Ensure username exists and password is correct
        if len(rows) != 1 or not check_password_hash(
            rows[0]["hash"], request.form.get("password")
        ):
            return apology("invalid username and/or password", 403)
        # Remember which user has logged in
        session["user_id"] = rows[0]["id"]
        # Redirect user to home page
        return redirect("/")
    # User reached route via GET (as by clicking a link or via redirect)
    else:
        return render_template("login.html")
@app.route("/logout")
def logout():
    """Log user out"""
    # Forget any user_id
    session.clear()
    # Redirect user to login form
```

```
    return redirect("/")
@app.route("/quote", methods=["GET", "POST"])
@login_required
def quote():
    """Get stock quote."""
    if request.method == "POST":
        symbol = request.form.get("symbol")
        # Validate input
        if not symbol:
            return apology("must provide symbol")
        stock = lookup(symbol)
        if stock is None:
            return apology("invalid symbol")
        return render_template("quoted.html", stock=stock)
    else:
        return render_template("quote.html")
@app.route("/register", methods=["GET", "POST"])
def register():
    """Register user"""
    if request.method == "POST":
        # Get form data
        username = request.form.get("username")
        password = request.form.get("password")
        confirmation = request.form.get("confirmation")
        # Validate input
        if not username:
            return apology("must provide username")
        if not password:
            return apology("must provide password")
        if not confirmation:
            return apology("must confirm password")
        if password != confirmation:
            return apology("passwords do not match")
        # Hash password before storing
        hash_ = generate_password_hash(password)
        # Insert new user into database
        try:
            db.execute("INSERT INTO users (username, hash) VALUES (?, ?)", username, hash_
)
        except ValueError:
            return apology("username already exists")
        rows = db.execute("SELECT id FROM users WHERE username = ?", username)
        # Automatically log in the new user
        session["user_id"] = rows[0]["id"]
        return redirect("/")
    else:
        return render_template("register.html")
@app.route("/sell", methods=["GET", "POST"])
@login_required
def sell():
    """Sell shares of stock"""
    user_id = session["user_id"]
    # Get user's current holdings
    rows = db.execute(
        """
        SELECT symbol, SUM(shares) AS shares
        FROM transactions
```

```
        WHERE user_id = ?
        GROUP BY symbol
        HAVING shares > 0
        """,
        user_id
    )
    symbols = [row["symbol"] for row in rows]
    if request.method == "POST":
        symbol = request.form.get("symbol")
        shares = request.form.get("shares")
        # Validate input
        if not symbol:
            return apology("must select symbol")
        if symbol not in symbols:
            return apology("invalid symbol")
        # Validate shares
        if not shares:
            return apology("must provide shares")
        try:
            shares = int(shares)
        except ValueError:
            return apology("shares must be positive integer")
        if shares <= 0:
            return apology("shares must be positive integer")
        # Get owned shares
        owned = db.execute(
            """
            SELECT SUM(shares) AS shares
            FROM transactions
            WHERE user_id = ? AND symbol = ?
            """,
            user_id, symbol
        )[0]["shares"]
        # Ensure user has enough shares to sell
        if shares > owned:
            return apology("too many shares")
        stock = lookup(symbol)
        price = stock["price"]
        total_value = shares * price
        # Update user's cash
        db.execute(
            "UPDATE users SET cash = cash + ? WHERE id = ?",
            total_value, user_id
        )
        # Record sell transaction (negative shares)
        db.execute(
            """
            INSERT INTO transactions (user_id, symbol, shares, price, type)
            VALUES (?, ?, ?, ?, 'SELL')
            """,
            user_id, symbol, -shares, price
        )
        return redirect("/")
    else:
        return render_template("sell.html", symbols=symbols)
@app.route("/password", methods=["GET", "POST"])
@login_required
```

```
def password():
    """Change password"""
    user_id = session["user_id"]
    if request.method == "POST":
        # Get form data
        old = request.form.get("old_password")
        new = request.form.get("new_password")
        confirmation = request.form.get("confirmation")
        # Validate input
        if not old:
            return apology("must provide current password")
        if not new:
            return apology("must provide new password")
        if not confirmation:
            return apology("must confirm new password")
        if new != confirmation:
            return apology("new passwords do not match")
        # Get current hash
        row = db.execute(
            "SELECT hash FROM users WHERE id = ?",
            user_id
        )
        # Verify current password
        if not check_password_hash(row[0]["hash"], old):
            return apology("current password incorrect")
        # Update to new password
        new_hash = generate_password_hash(new)
        db.execute(
            "UPDATE users SET hash = ? WHERE id = ?",
            new_hash, user_id
        )
        flash("Password changed successfully")
        return redirect("/")
    else:
        return render_template("change_password.html")
```

Helpers

```
import requests
# Flask utilities for redirection, template rendering, and session handling
from flask import redirect, render_template, session
# Utility to preserve function metadata when creating decorators
from functools import wraps
def apology(message, code=400):
    """Render message as an apology to user."""
    def escape(s):
        """
        Escape special characters.
        Reference:
        https://github.com/jacebrowning/memegen#special-characters
        """
        for old, new in [
            ("-", "--"),
            (" ", "-"),
            ("_", "__"),
```

```
            ("?", "~q"),
            ("%", "~p"),
            ("#", "~h"),
            ("/", "~s"),
            ('"', "''"),
        ]:
            s = s.replace(old, new)
        return s
    # Render apology template with escaped message and status code
    return render_template(
        "apology.html",
        top=code,
        bottom=escape(message)
    ), code
def login_required(f):
    """
    Decorator that requires a user to be logged in
    before accessing the decorated route.
    Reference:
    https://flask.palletsprojects.com/en/latest/patterns/viewdecorators/
    """
    @wraps(f)
    def decorated_function(*args, **kwargs):
        # Check if user is logged in
        if session.get("user_id") is None:
            return redirect("/login")
        return f(*args, **kwargs)
    # Execute the original function
    return decorated_function
def lookup(symbol):
    """Look up quote for symbol."""
    # Build API request URL using the provided symbol
    url = f"https://finance.cs50.io/quote?symbol={symbol.upper()}"
    try:
        # Send HTTP request to CS50 Finance API
        response = requests.get(url)
        # Raise an error for HTTP error responses
        response.raise_for_status()
        # Parse JSON response
        quote_data = response.json()
        # Return structured quote information
        return {
            "name": quote_data["companyName"],
            "price": quote_data["latestPrice"],
            "symbol": symbol.upper()
        }
    except requests.RequestException as e:
        # Handle network-related errors
        print(f"Request error: {e}")
    except (KeyError, ValueError) as e:
        # Handle unexpected or malformed API responses
        print(f"Data parsing error: {e}")
    return None
def usd(value):
    """Format value as USD."""
    return f"${value:,.2f}"
```

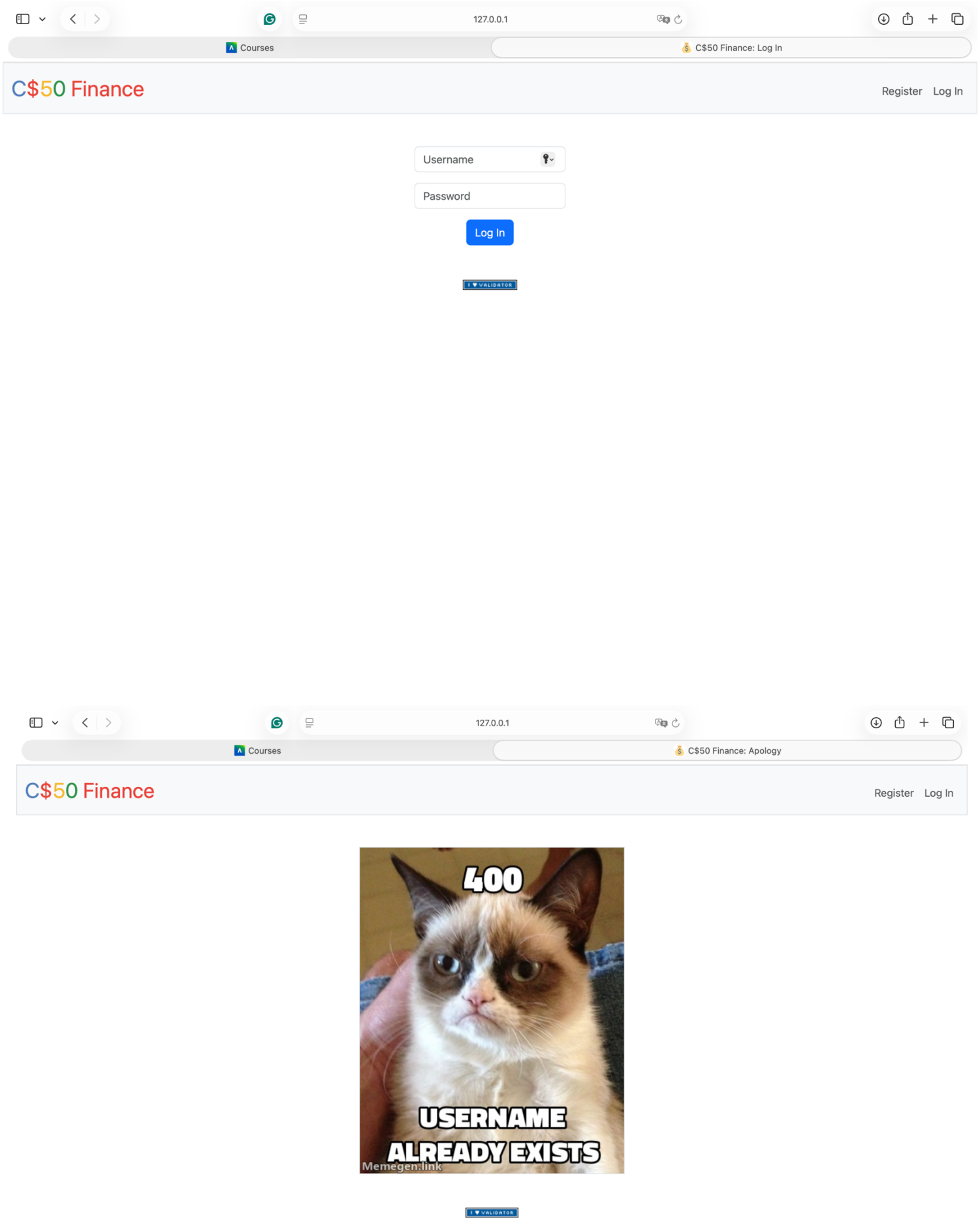


Figure 23 - Website | Register

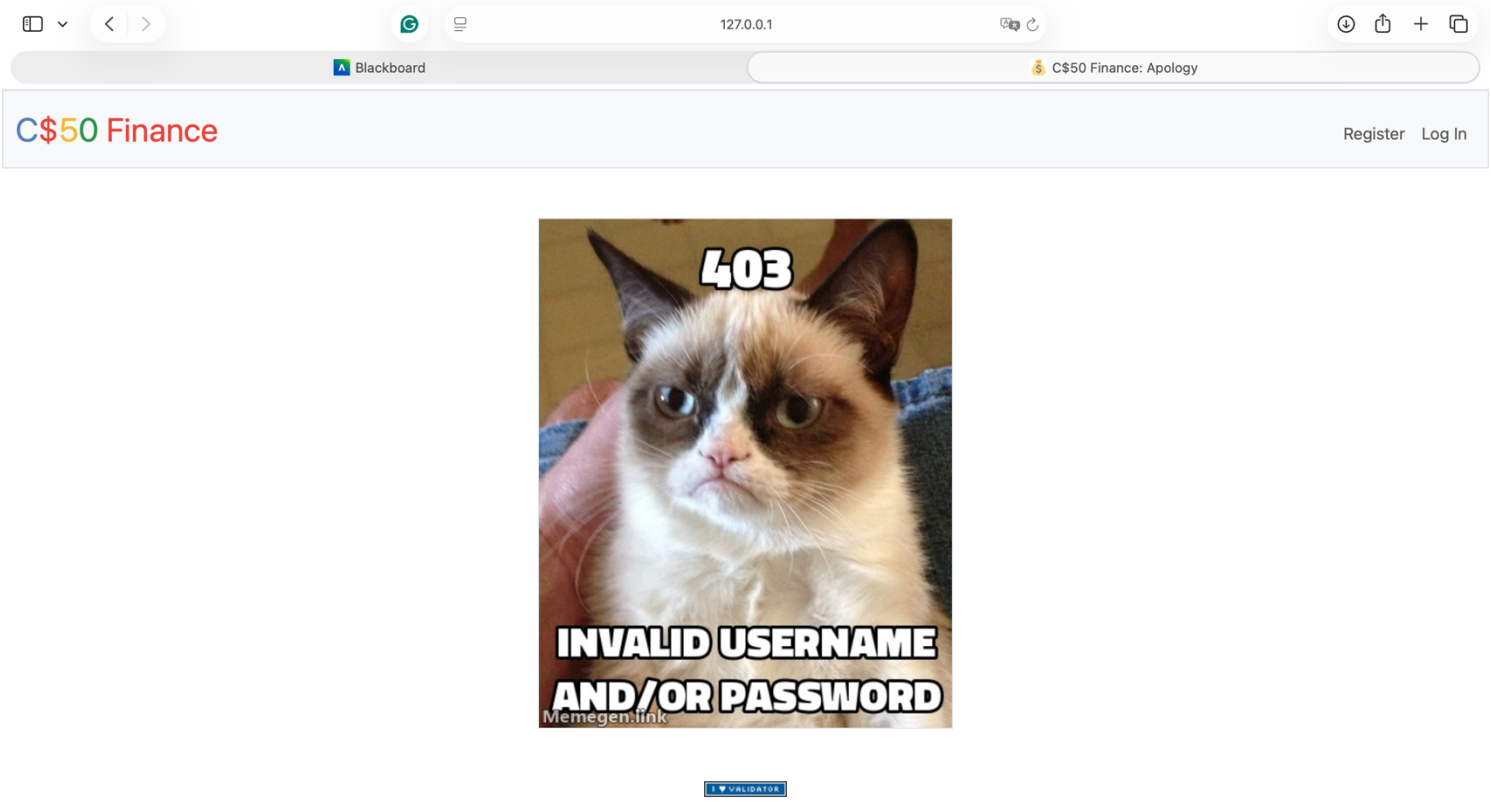
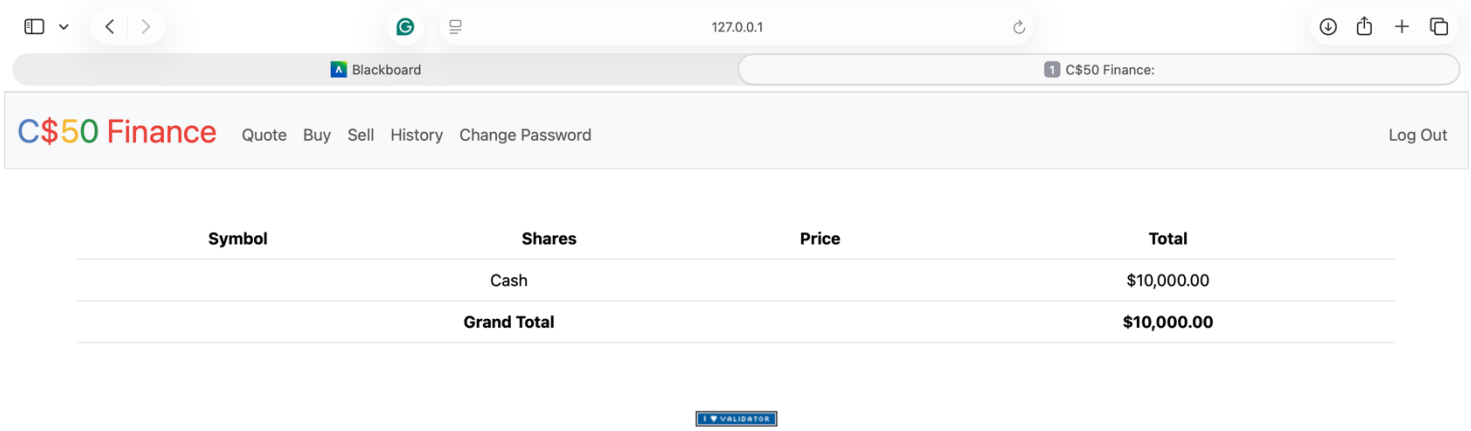


Figure 24 - Website | User_Error



Symbol	Shares	Price	Total
	Cash		$10,000.00
	Grand Total		**$10,000.00**

Figure 25 - Website | Password_Error

Figure 26 - Website | Finance

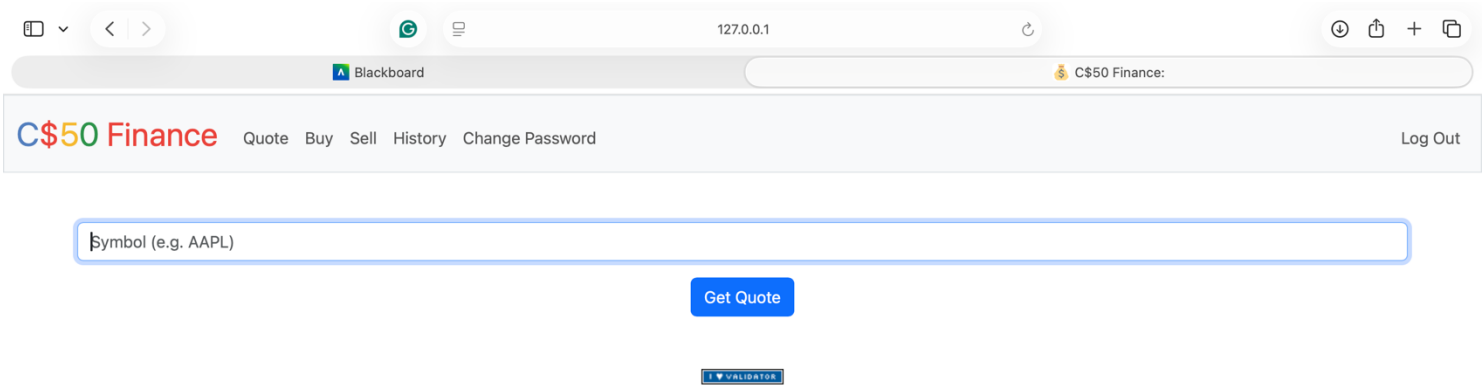


Figure 27 - Website | Quote

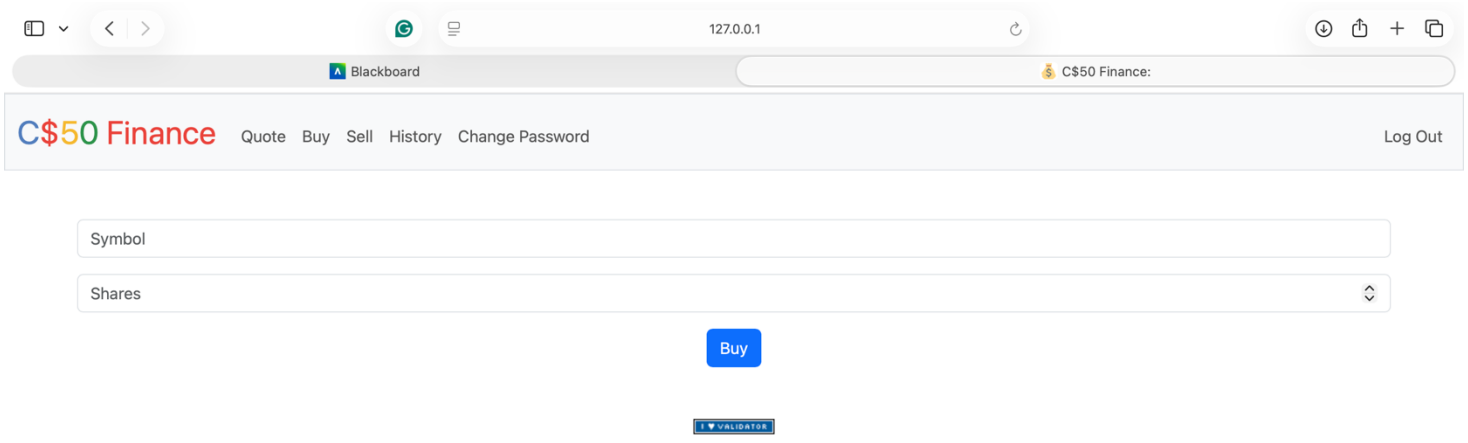


Figure 28 - Website | Buy

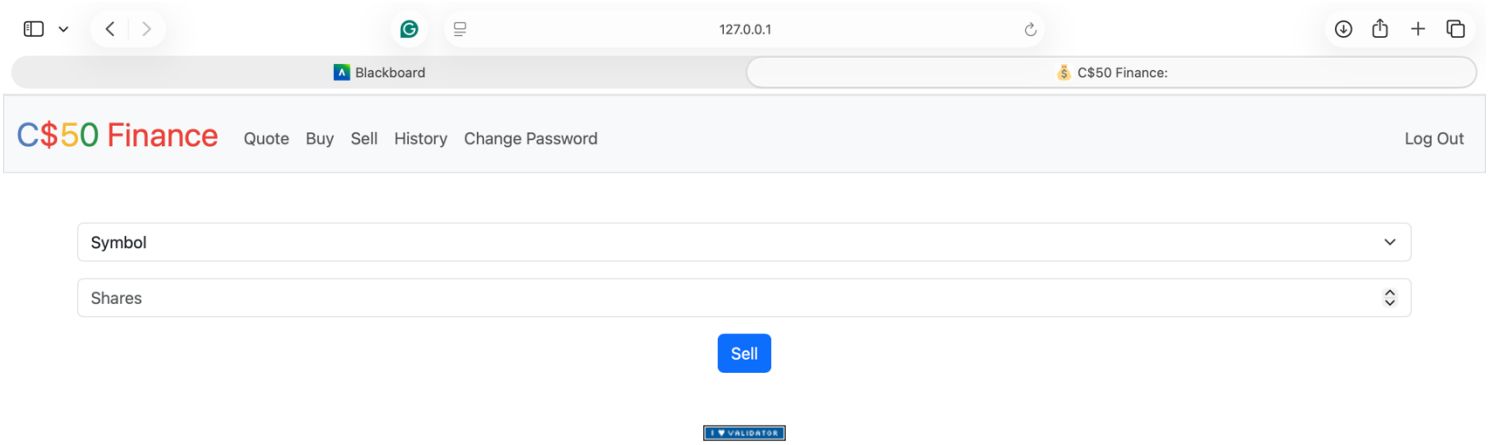


Figure 29 - Website | Sell

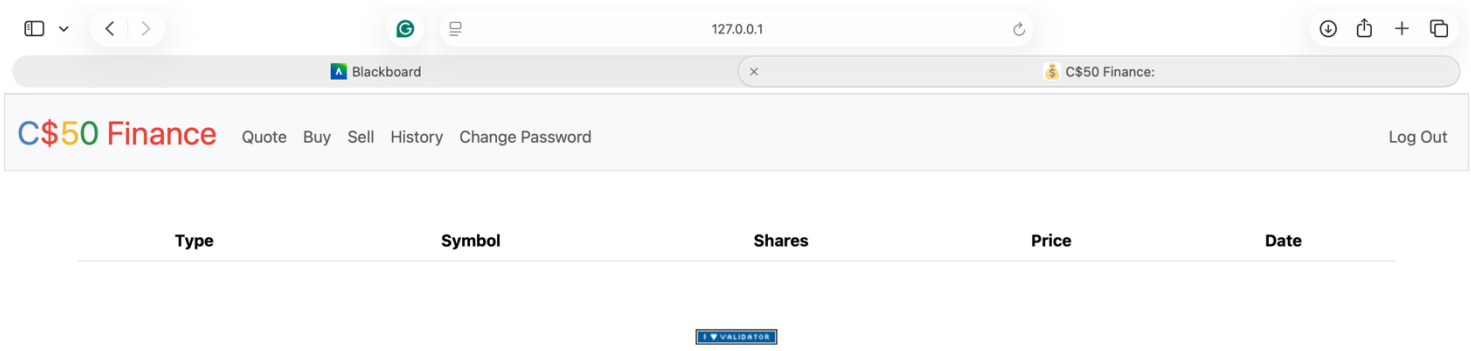


Figure 30 - Website | History

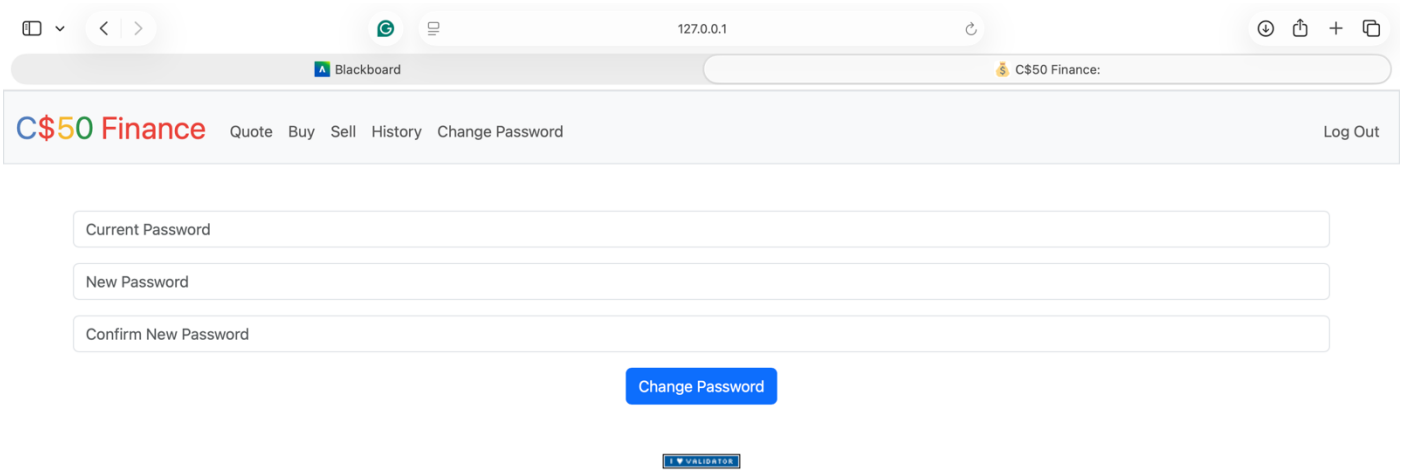


Figure 31 - Website | Change Password

me50 / users / franciscpime / cs50 / problems / 2025 / x / finance

#1 submitted a few seconds ago, Tuesday, December 30, 2025 2:41 PM WET
check50 21/21 • style50 1.00 • 0 comments
tar.gz • zip

Week 5

Show me your VENV

This project demonstrates the complete process of developing a **Flask web application** to **Render**, turning a locally developed app into a publicly accessible web service. The deployment follows standard industry practices, including virtual environments, dependency management, version control with git and cloud deployment.

1. Local Setup and Environment Configuration

The application was developed using Visual Studio Code, with a Python virtual environment to ensure dependency isolation and reproducibility.

A virtual environment was created and activated, followed by the installation of the required packages. Flask is used as the web framework, while Gunicorn is used as the production WSGI server.

```
python -m venv venv
source venv/bin/activate
pip install flask gunicorn
pip freeze > requirements.txt
```

The `requirements.txt` file ensures that all dependencies are correctly installed during deployment.

2. Project Structure and Render Configuration

To be compatible with render, the project follows a clean and minimal structure. The root directory contains:

- `app.py` - main flask application file;
- `requirements.txt` - Python dependencies;
- `Procfile` - startup instructions for render.

The `Procfile` defines how the application should be launched in production:

```
web: gunicorn app:app
```

This configuration assumes that the flask application instance is named `app` inside the `app.py` file.

A `.gitignore` file is also included to prevent unnecessary files (such has the virtual environment and cache files) from being tracked by Git.

3. Version Control and GitHub Integration

The project is version-controlled using **Git** and hosted on **GitHub** which allows Render to automatically deploy the application from the repository.

After initializing the repository locally, the code is committed and pushed to a remote GitHub repository:

```
git init
git add .
git commit -m "Initial commit"
git push -u origin master
```

4. Deployment on Render

The deployment is handled directly through the Randall platform:

- A new **Web Service** is created on Render;
- The GitHub repository is connected;
- Render automatically installs the dependencies and builds the application.

Once the build process is complete, render provides a **public URL** where the application is live.

5. Testing and Debugging

After deployment, the application is tested using the generated URL.

Common issues during deployment typically include missing dependencies or incorrect configuration. These are resolved by:

- Ensuring the `requirements.txt` file is up to date;
- Verifying that the flask app object name matches the one specified in the `Procfile`;
- Confirming that `app.py` is located in the root directory.

Arduino MAC-Address

In this project, each Arduino device needed to be uniquely identified on the network. To achieve this, I retrieved the **MAC address** of my Arduino after establishing a Wi-Fi connection. The MAC address serves as a unique hardware identifier, which is particularly useful for device identification, logging, or database association.

1. WiFi Connection Setup

The Wi-Fi connection logic is encapsulated in a dedicated module (`connect.h` / `connect.cpp`) to keep the code base clean and modular.

The header filed declares all required external variables and the `connect()` function, including:

- Wi-Fi credentials;
- Server configuration;
- Network and http client instances.

This separation allows the connection logic to be reused and maintained independently.

2. Establishing the Connection

Inside the `connect()` function, the Arduino continuously attempts to connect to the specified Wi-Fi network until the connection is successful:

```
while (status != WL_CONNECTED) {
  Serial.print("Connecting to ");
  Serial.println(ssid);
  status = WiFi.begin(ssid, pass);
}
```

Once connected, the device prints confirmation messages and its assigned local IP address to the Serial Monitor, ensuring that the network connection is active and stable.

3. Retrieving the MAC Address

After the Wi-Fi connection is established, the MAC address is obtained using the built-in Wi-Fi library:

```
byte mac[6];
WiFi.macAddress(mac);
```

The MAC address is stored as a 6-byte array, where each byte represents one hexadecimal segment of the address.

4. Formatting and Displaying the MAC Address

To make the MAC address human-readable and consistent with standard formatting, each byte is printed in hexadecimal format, separated by colons:

```
for (int i = 0; i < 6; i++) {
  if (mac[i] < 16)
    Serial.print(0);
  Serial.print(mac[i], HEX);
  if (i < 5)
    Serial.print(':');
}
```

Leading zeros are added when necessary to ensure that every segment contains 2 hexadecimal digits. The final result follows the conventional format:

```
XX:XX:XX:XX:XX:XX
```

This output is displayed directly in the Serial Monitor, allowing each user to easily identify and record the MAC address of their Arduino board.

Code

connect.h

```
#pragma once
#include <SPI.h>
#include <WiFi.h>
#include <ArduinoHttpClient.h>
#include "wifi_secrets.h"
extern char ssid[];
extern char pass[];
extern int status;
extern char serverAddress[];
extern int port;
//  WiFiClient instance
extern WiFiClient wifi;
// HttpClient instance:
extern HttpClient httpClient;
void connect();
```

connect.cpp

```
#include "connect.h"
// global
char ssid[] = SECRET_SSID;
char pass[] = SECRET_PASS;
int status = WL_IDLE_STATUS;
char serverAddress[] = DB_URL;
int port = DB_PORT;
//  WiFiClient instance
WiFiClient wifi;
// HttpClient instance:
HttpClient httpClient = HttpClient(wifi, serverAddress, port);
void connect() {
  while (status != WL_CONNECTED) {
    Serial.print("Connecting to ");
    Serial.println(ssid);
    status = WiFi.begin(ssid, pass);
  }
  delay(4000);
  byte mac[6];
  Serial.println("WiFi connected!");
  Serial.print("IP address: ");
  Serial.println(WiFi.localIP());
  WiFi.macAddress(mac);
  for (int i = 0; i < 6; i++) {
    if (mac[i] < 16)
      Serial.print(0);
    Serial.print(mac[i], HEX);
    if (i < 5)
      Serial.print(':');
  }
  Serial.println("");
}
```

Embedded Systrems

Week 0

Interview 1

```
//---------------------------------------------------------------------
// Word Lottery on LCD Display
// ===========================================
//
// Displays two randomly changing words on a 16x2 I2C LCD.
// Each line shows a word from a different list.
// When the button is pressed, the display pauses on the current words.
// Pressing the button again resumes the random word rotation.
//
// Features:
// - Two independent word lists (CS50 and Embedded Systems).
// - I2C LCD (16x2) for compact and clear display.
// - Button toggle to start/stop random rotation.
// - Simple and intuitive user interaction.
//
// Francisco Pimentel
// 08 / Nov / 2025
// version 2.3.6
//---------------------------------------------------------------------
#include <LCD_I2C.h>                  // Include LCD I2C library
#define BUTTON_PIN 7                  // Define button pin
LCD_I2C lcd(0x27, 16, 2);             // Create LCD object (I2C address 0x27, 16x2 display)
bool isRunning       = true;          // Keeps track if the words are rotating
bool lastButtonState = HIGH;          // Store last button state for toggle detection
bool button;
// CS50 Words
String line1[] = {
  "Scratch", "Mario Less", "Mario More", "Cash", "Credit", "Scrabble",
  "Readability", "Caesar", "Substitution", "Plurality", "Runoff",
  "Tideman", "Filter Less", "Filter More", "Recover", "Git Together",
  "Inheritance", "Speller"
};
// Embedded Systems Words
String line2[] = {
  "CuteBot", "Mario UNO", "Morse Code",
  "Traffic Lights", "Uptimer", "Jumbo Clock",
  "Weather Station"
};
const int wordsInLine1 = sizeof(line1) / sizeof(line1[0]);      // Count of words in line1
const int wordsInLine2 = sizeof(line2) / sizeof(line2[0]);      // Count of words in line2
void setup() {
  Serial.begin(9600);                                           // Start serial communication
  lcd.begin();                                                  // Initialize LCD
  lcd.backlight();                                              // Turn on LCD backlight
  pinMode(BUTTON_PIN, INPUT_PULLUP);      // Set button pin as input with internal pull-up
  randomSeed(analogRead(A0));             // Initialize random seed from analog pin A0
}
void loop() {
```

```
  button = digitalRead(BUTTON_PIN);                    // Read current button state
  // Detect button press (HIGH >> LOW)
  if (lastButtonState == HIGH && button == LOW) {
    isRunning = !isRunning;                            // Toggle rotation on/off
    delay(250);                                        // Debounce delay to avoid multiple toggles
  }
  lastButtonState = button;                            // Update last button state
  if (isRunning) {                                     // If rotation is active
    lcd.clear();                                       // Clear LCD
    lcd.setCursor(0, 0);                               // Set cursor to first line
    lcd.print(line1[random(wordsInLine1)]);            // Print random word from line1
    lcd.setCursor(0, 1);                               // Set cursor to second line
    lcd.print(line2[random(wordsInLine2)]);            // Print random word from line2
    delay(150);                                        // Short delay before next rotation
  }
```

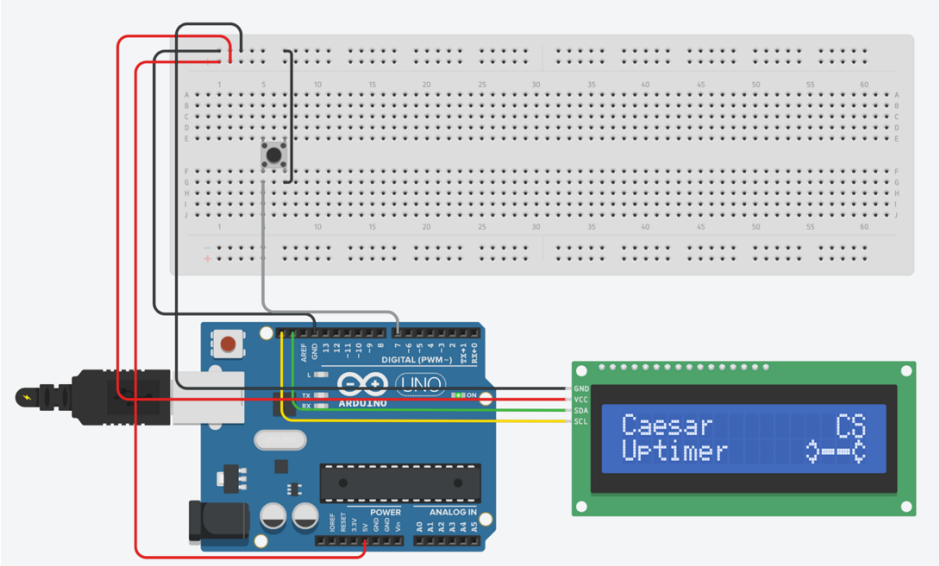


Figure 32 - Interview1 | Circuit View

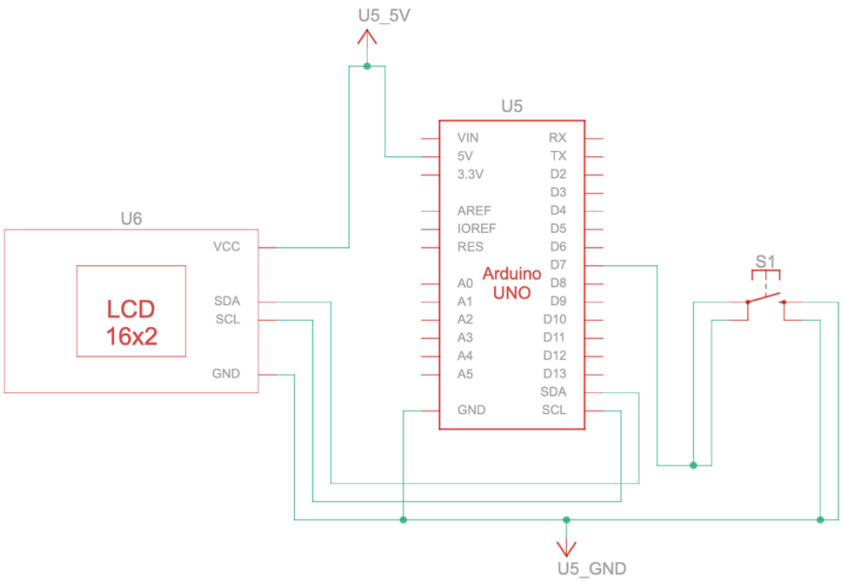


Figure 33 - Interview1 | Schematic View

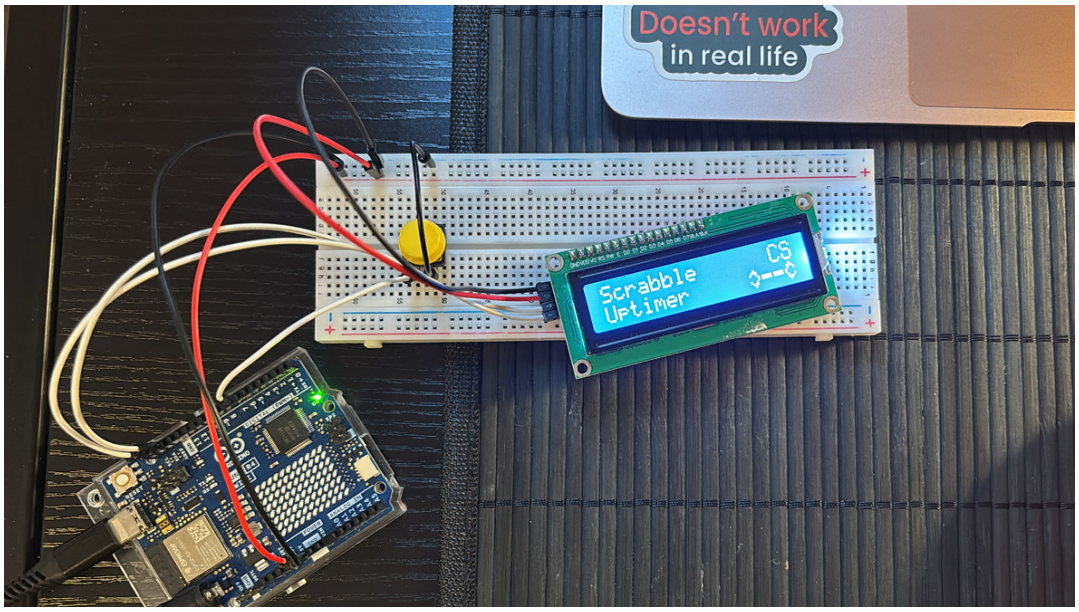


Figure 34 - Interview1 | Real View

Access link: https://www.tinkercad.com/things/1kII47U4M91-interview1?sharecode=sV3mG6fe49oluLoxrQOwaieiIaWvkd_ejrTd5HLud0o

Project Debriefing - Retro2CBot

Project Overview

Retro2CBot was a team-based robotics project focused on designing and implementing an autonomous robot capable of navigating multiple maze environments. The robot was required to solve line following tracks, line maze and physical maze while operating fully autonomously and transmitting real-time data to an external monitoring system.

The project combined hardware design, embedded software development, system architecture, and communication systems, making it a comprehensive engineering challenge.

My Role and Contributions

Within the team, I was deeply involved in the software architecture and system organization of the project. My main contributions included:

- Structuring the project into modular, reusable software components;
- Enforcing a clear separation between hardware control, behavior logic, and communication;
- Contributing to navigation logic design and non blocking execution strategies;

Beyond implementation, I played a key role in organizing and refining the technical documentation, ensuring that the final report clearly reflected the engineering decisions behind the system.

Technical Challenges

One of the main challenges was managing complexity. As the project grew, it became clear that a monolithic approach would quickly become and unmaintainable. Designing a scalable architecture that could support multiple robot boards and navigation strategies required careful planning and interaction.

Another challenge was ensuring real-time responsiveness. Replacing blocking delays with non-blocking logic based on timers was essential to maintain continuous sensors updates, stable navigation and reliable communication.

Integrating hardware feedback (such as motor pulse sensors) into control logic also require the solid understanding of embedded systems and timing constraints.

What I Learned

This project helped me with:

- Modular software design in embedded systems;
- Hardware abstractions and code reuse;
- Non-blocking programming and time based control;
- Translating sensor data into high-level navigation decisions;
- System-level thinking, beyond individual components.

I also learned how critical clear documentation and structure are in collaborative technical projects, both for development and evaluation.

Self-Reflection

This period was a time of significant learning and personal growth for me. Unlike in the past when my contact with programming was more limited and focused on a single language, this experience allowed me to explore and work with several different programming languages. Being exposed to multiple languages helped me better understand the logic behind programming, rather than just memorizing syntax. I learned how different languages approach problem-solving and how to adapt my thinking depending on the tools being used. This broadened my perspective and made me more confident as a learner and has a future professional in this field.

Another very important aspect of this period was the group project. Working in a team taught me lessons that go far beyond technical knowledge. I learned how to communicate my ideas more clearly, how to listen to others, and how to accept constructive criticism. There were moments of difficulty such as coordinating tasks and managing team parent opinions, but these challenges helped me develop teamwork and responsibility. I also realized the importance of planning, time management, and mutual support in order to achieve common goals.

Overall, this was a demanding but extremely rewarding period. I not only improved my technical skills, but also developed soft skills that are essential in any professional environment. Compared to my past experiences, I feel that I have grown significantly, both academically and personally, and I am more prepared to face future challenges with confidence and motivation.

P2. Computer Science

Name	Student nr.	Group	Year
Francisco Ponte Pimentel		Retro2Cbot	25/26

The student is responsible for safe keeping of this card

Assignments

Wk	Subject	Date	Signature
2.1	Hello, Mario Less or More, Cash or Credit	12 NOV 2025	
	Readability	12 NOV 2025	
	DNA	12 NOV 2025	
2.2	Songs Movies	12 NOV 2025	
	Fifty Ville	27 NOV 2025	
2.3	Trivia	8 DEC 2025	
	Homepage	8 DEC 2025	really good
	Hand-in Technical Design		
2.4	Birthdays	15 DEC 2025	
	Finance	15 DEC 2025	
2.5	Show me your VENV	15 DEC 2025	
	Arduino MAC-Address	12 DEC 2025	
2.6	Team website + Digital Twin		
2.8	Final project and completing portfolio		
T2	**Assessment**		

Feedback:

This card is proof that the above assignments are properly completed and should be:

- signed off by the appropriate teacher(s), in the same week, a week before, or a week after, during the guided ateliers
- a digital attachment of your portfolio and
- physically handed over at the beginning of the assessment

NHL STENDEN
university of applied sciences

P2. Embedded Systems

Name	Student nr.	Group	Year
Francisco Ruenes	5708915	Robo2CH	25/26

The student is responsible for safe keeping of this card

Assignments

Wk	Subject	Date	Signature
2.1	Basic movements *(Functions)*	21 NOV 2025	M. Baro
2.2	Object avoidance	25 NOV 2025	M. Baro
2.3	Stay on track *(Arrays)*	16 DEC 2025	M. Baro
2.4	**Midterm race** *(subgroup)*	~3 DEC 2025	Bart
2.5	Connectivity *(HC12)*	16 DEC 2025	M. Baro
2.6	Data storage *(SQL)*	16 DEC 2025	M. Baro
2.7	**Test Day** + Demo Dashboard *(team)*	~5 JAN 2026	Bart
2.8	**Race Day** *(Circuit Challenge)* **+ Project Dossier**		
T2	**Assessment**		

Feedback:

This card is proof that the above assignments are properly completed and should be:

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